

Appendix A: Salary component data package

Field guide and usage notes

Table A1: Field guide and usage notes

Field	Explanation
Monthly-equivalent Low / High / Midpoint	Monthly salary, hourly wage, daily wage and 13th-month-pay postings are converted into a comparable monthly range.
Estimated Annual Cash Wage	Monthly-equivalent midpoint $\times$ salary months; if 13th-month pay is not explicitly stated, 12 months is used.
Estimated Annual Employer Labour Cost	Estimated by adding an assumed 30% employer burden to cash wages; used in the robot-versus-human labour-cost model.
Salary Component Fields	Night shift, meal allowance, accommodation, social insurance, housing fund, overtime, piece-rate/performance pay, bonus/allowance/13th-month pay and similar items are separated into independent columns.
Reliability Notes	Based on public recruitment pages or listing excerpts. Pages are dynamic; secondary verification is recommended before formal economic modelling.
Source	Original content comes from the pasted text table uploaded by the user; source URLs are retained in the workbook.

Brand-level salary summary

Table A2: Brand-level salary summary

Brand	Samples	Avg monthly low	Avg monthly high	Avg monthly midpoint	Avg employer labour cost
JD Logistics	68	RMB 5,946	RMB 7,622	RMB 6,784	RMB 105,971
SF Express	30	RMB 5,503	RMB 7,357	RMB 6,430	RMB 100,311
Total	98	RMB 5,813	RMB 7,543	RMB 6,678	RMB 104,279

Brand-level salary component shares

Table A3: Brand-level salary component shares

Brand	Night shift	Direct hire	Social insurance	Housing fund	Accommodation	Meal allowance	Overtime	Piece/performance	Bonus/13th
JD Logistics	38.2%	52.9%	35.3%	30.9%	14.7%	11.8%	4.4%	7.4%	16.2%
SF Express	50.0%	83.3%	10.0%	0.0%	16.7%	13.3%	6.7%	6.7%	10.0%
Total	41.8%	62.2%	27.6%	21.4%	15.3%	12.2%	5.1%	7.1%	14.3%

City-level salary summary

Table A4: City-level salary summary

Brand	City	N	Avg low	Avg high	Avg midpoint	Avg employer cost	Representative jobs
JD Logistics	Shanghai	5	RMB 6,800	RMB 7,800	RMB 7,300	RMB 113,880	Parcel Sorter; Part-time Parcel Sorter; Sorting and Stock Handling Worker
JD Logistics	Zhongshan	1	RMB 6,000	RMB 7,000	RMB 6,500	RMB 101,400	Parcel Sorter
JD Logistics	Urumqi	1	RMB 5,000	RMB 7,000	RMB 6,000	RMB 93,600	Parcel Sorter
JD Logistics	Beijing	3	RMB 5,667	RMB 6,667	RMB 6,167	RMB 96,200	Store/Front-warehouse Picking and Packing Worker; Direct-hire Temporary/short-term Parcel Sorter; Direct-hire Store/Front-warehouse Picking and Packing Worker
JD Logistics	Nanjing	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	High-pay Warehouse Operator
JD Logistics	Nanning	2	RMB 5,100	RMB 6,750	RMB 5,925	RMB 92,430	Warehouse Worker; Picking and Packing Warehouse Worker
JD Logistics	Nanchang	2	RMB 5,500	RMB 7,000	RMB 6,250	RMB 97,500	Temporary/short-term Sorting and Stock Handling Worker; Direct-hire Warehouse Operator
JD Logistics	Nantong	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Parcel Sorter
JD Logistics	Hefei	3	RMB 5,733	RMB 7,500	RMB 6,617	RMB 103,220	Parcel Sorter; Direct-hire Warehouse Worker; Direct-hire Part-time Parcel Sorter
JD Logistics	Dalian	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Parcel Sorter
JD Logistics	Tianjin	3	RMB 5,900	RMB 7,433	RMB 6,667	RMB 104,000	Dispatch, Stock Handling, Picking and Shipping Worker; Direct-hire Part-time High-pay Parcel Sorter; Part-time Parcel Sorter
JD Logistics	Taiyuan	1	RMB 5,200	RMB 6,500	RMB 5,850	RMB 91,260	Part-time Sorting and Loading/Unloading Worker
JD Logistics	Ningbo	2	RMB 6,500	RMB 8,000	RMB 7,250	RMB 113,100	Parcel Sorter; Sorting and Loading/Unloading Worker
JD Logistics	Guangzhou	6	RMB 6,500	RMB 8,667	RMB 7,583	RMB 119,925	Parcel Sorter; Parcel Sorter; Direct-hire Regular Warehouse Operator
JD Logistics	Yanbian	1	RMB 4,000	RMB 5,000	RMB 4,500	RMB 70,200	Parcel Sorter
JD Logistics	Xuzhou	1	RMB 5,200	RMB 6,500	RMB 5,850	RMB 91,260	Direct-hire Part-time Parcel Sorter
JD Logistics	Chengdu	6	RMB 5,367	RMB 7,250	RMB 6,308	RMB 98,410	Direct-hire Warehouse Worker; Warehouse Worker; Part-time Parcel Sorter

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Brand	City	N	Avg low	Avg high	Avg midpoint	Avg employer cost	Representative jobs
JD Logistics	Wuxi	3	RMB 6,667	RMB 8,000	RMB 7,333	RMB 114,400	Regular Parcel Sorter; Picking and Packing Warehouse Worker; Direct-hire Sorting and Loading/Unloading Worker
JD Logistics	Hangzhou	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Direct-hire Regular Parcel Sorter
JD Logistics	Wuhan	5	RMB 5,440	RMB 7,700	RMB 6,570	RMB 102,492	Parcel Sorter; Direct-hire Picking and Packing Warehouse Worker; Sorting and Stock Handling Worker
JD Logistics	Shenyang	1	RMB 8,000	RMB 10,000	RMB 9,000	RMB 140,400	Warehouse Operator
JD Logistics	Cangzhou	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Parcel Sorter
JD Logistics	Quanzhou	1	RMB 7,000	RMB 10,000	RMB 8,500	RMB 132,600	Regular Parcel Sorter
JD Logistics	Taizhou	1	RMB 5,200	RMB 6,500	RMB 5,850	RMB 91,260	Dispatch, Stock Handling, Picking and Shipping Worker
JD Logistics	Luoyang	1	RMB 5,000	RMB 6,000	RMB 5,500	RMB 85,800	Direct-hire Parcel Sorter
JD Logistics	Zibo	1	RMB 6,000	RMB 7,000	RMB 6,500	RMB 101,400	Direct-hire Parcel Sorter
JD Logistics	Wenzhou	1	RMB 5,200	RMB 6,500	RMB 5,850	RMB 91,260	Direct-hire Part-time Parcel Sorter
JD Logistics	Zhuhai	2	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Direct-hire Regular Warehouse Worker; Parcel Sorter
JD Logistics	Fuzhou	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Sorting and Stock Handling Worker
JD Logistics	Wuhu	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Sorting and Loading/Unloading Worker
JD Logistics	Suzhou	2	RMB 6,500	RMB 8,000	RMB 7,250	RMB 113,100	Regular Sorting and Loading/Unloading Worker; Temporary/short-term Regular Picking and Packing Warehouse Worker
JD Logistics	Jingzhou	1	RMB 6,000	RMB 7,000	RMB 6,500	RMB 101,400	Sorting and Loading/Unloading Worker
JD Logistics	Guiyang	1	RMB 5,000	RMB 9,000	RMB 7,000	RMB 109,200	Direct-hire Sorting and Loading/Unloading Worker
JD Logistics	Xingtai	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Parcel Sorter
JD Logistics	Zhengzhou	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Direct-hire Regular Sorting and Loading/Unloading Worker
JD Logistics	Chongqing	1	RMB 5,000	RMB 6,000	RMB 5,500	RMB 85,800	Dispatch, Stock Handling, Picking and Shipping Worker
JD Logistics	Zhenjiang	1	RMB 6,000	RMB 9,000	RMB 7,500	RMB 117,000	Regular Warehouse Worker
SF Express	Sanya	1	RMB 5,000	RMB 6,000	RMB 5,500	RMB 85,800	Parcel Sorter
SF Express	Shanghai	1	RMB 6,000	RMB 9,000	RMB 7,500	RMB 117,000	Parcel Sorter
SF Express	Dongguan	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Sorting/Forklift Worker; Parcel Sorter
SF Express	Zhongshan	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Direct-hire Parcel Sorter
SF Express	Beijing	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Parcel Sorter
SF Express	Nanjing	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Parcel Sorter
SF Express	Nanping	1	RMB 6,000	RMB 7,000	RMB 6,500	RMB 101,400	Sorting and Loading/Unloading Worker
SF Express	Nanchang	1	RMB 5,200	RMB 10,400	RMB 7,800	RMB 121,680	Temporary/short-term Sorting/Forklift Worker
SF Express	Xiamen	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Parcel Sorter
SF Express	Hefei	1	RMB 5,200	RMB 6,500	RMB 5,850	RMB 91,260	Part-time Parcel Sorter; Parcel Sorter
SF Express	Dalian	1	RMB 5,000	RMB 7,000	RMB 6,000	RMB 93,600	Part-time Parcel Sorter
SF Express	Tianjin	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Direct-hire Parcel Sorter
SF Express	Taiyuan	1	RMB 5,000	RMB 7,000	RMB 6,000	RMB 93,600	Parcel Sorter
SF Express	Ningbo	1	RMB 7,000	RMB 9,000	RMB 8,000	RMB 124,800	Parcel Sorter
SF Express	Guangzhou	1	RMB 5,000	RMB 8,000	RMB 6,500	RMB 101,400	Parcel Sorter
SF Express	Chengdu	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Dispatch, Stock Handling, Picking and Shipping Worker
SF Express	Hangzhou	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	High-pay Parcel Sorter
SF Express	Guilin	1	RMB 4,000	RMB 7,000	RMB 5,500	RMB 85,800	Sorting and Loading/Unloading Worker
SF Express	Wuhan	1	RMB 6,000	RMB 7,000	RMB 6,500	RMB 101,400	Direct-hire Parcel Sorter
SF Express	Cangzhou	1	RMB 5,000	RMB 6,000	RMB 5,500	RMB 85,800	Parcel Sorter
SF Express	Jinan	1	RMB 5,000	RMB 8,000	RMB 6,500	RMB 101,400	Sorting and Loading/Unloading Worker
SF Express	Haikou	1	RMB 5,000	RMB 7,000	RMB 6,000	RMB 93,600	Parcel Sorter
SF Express	Shenzhen	1	RMB 7,000	RMB 8,000	RMB 7,500	RMB 117,000	Parcel Sorter
SF Express	Welfang	1	RMB 2,600	RMB 3,250	RMB 2,925	RMB 45,630	Part-time Parcel Sorter
SF Express	Suzhou	1	RMB 0	RMB 0	RMB 0	RMB 0	To be added
SF Express	Jingzhou	1	RMB 2,600	RMB 5,200	RMB 3,900	RMB 60,840	Part-time Sorting and Loading/Unloading Worker
SF Express	Yingkou	1	RMB 4,000	RMB 6,000	RMB 5,000	RMB 78,000	Parcel Sorter
SF Express	Xi'an	1	RMB 6,000	RMB 8,000	RMB 7,000	RMB 109,200	Parcel Sorter
SF Express	Xingtai	1	RMB 5,000	RMB 7,000	RMB 6,000	RMB 93,600	Parcel Sorter; Parcel Sorter
SF Express	Changchun	1	RMB 6,000	RMB 7,000	RMB 6,500	RMB 101,400	Parcel Sorter

City-level salary component shares

Table A5: City-level salary component shares

Brand	City	N	Night shift	Social insurance	Accommodation	Meal allowance	Piece/performance
JD Logistics	Shanghai	5	20.0%	60.0%	40.0%	0.0%	0.0%
JD Logistics	Zhongshan	1	0.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Urumqi	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Beijing	3	33.3%	33.3%	0.0%	0.0%	0.0%
JD Logistics	Nanjing	1	0.0%	0.0%	0.0%	0.0%	100.0%
JD Logistics	Nanning	2	50.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Nanchang	2	50.0%	50.0%	50.0%	50.0%	0.0%
JD Logistics	Nantong	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Hefei	3	66.7%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Dalian	1	100.0%	100.0%	0.0%	0.0%	0.0%
JD Logistics	Tianjin	3	66.7%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Taiyuan	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Ningbo	2	0.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Guangzhou	6	0.0%	66.7%	16.7%	33.3%	0.0%
JD Logistics	Yanbian	1	0.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Xuzhou	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Chengdu	6	16.7%	33.3%	0.0%	0.0%	0.0%
JD Logistics	Wuxi	3	33.3%	0.0%	0.0%	33.3%	66.7%
JD Logistics	Hangzhou	1	0.0%	100.0%	100.0%	100.0%	0.0%
JD Logistics	Wuhan	5	20.0%	40.0%	0.0%	20.0%	0.0%
JD Logistics	Shenyang	1	0.0%	0.0%	100.0%	0.0%	0.0%
JD Logistics	Cangzhou	1	0.0%	100.0%	0.0%	100.0%	0.0%
JD Logistics	Quanzhou	1	0.0%	100.0%	100.0%	0.0%	0.0%
JD Logistics	Taizhou	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Luoyang	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Zibo	1	100.0%	100.0%	0.0%	0.0%	0.0%
JD Logistics	Wenzhou	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Zhuhai	2	0.0%	50.0%	100.0%	0.0%	0.0%
JD Logistics	Fuzhou	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Wuhu	1	100.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Suzhou	2	50.0%	50.0%	0.0%	0.0%	50.0%
JD Logistics	Jingzhou	1	100.0%	100.0%	0.0%	0.0%	0.0%

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Brand	City	N	Night shift	Social insurance	Accommodation	Meal allowance	Piece/performance
JD Logistics	Guiyang	1	0.0%	100.0%	0.0%	0.0%	0.0%
JD Logistics	Xingtai	1	100.0%	0.0%	100.0%	100.0%	100.0%
JD Logistics	Zhengzhou	1	100.0%	100.0%	0.0%	0.0%	0.0%
JD Logistics	Chongqing	1	0.0%	0.0%	0.0%	0.0%	0.0%
JD Logistics	Zhenjiang	1	0.0%	100.0%	0.0%	0.0%	0.0%
SF Express	Sanya	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Shanghai	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Dongguan	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Zhongshan	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Beijing	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Nanjing	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Nanping	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Nanchang	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Xiamen	1	100.0%	100.0%	100.0%	0.0%	100.0%
SF Express	Hefei	1	100.0%	100.0%	100.0%	100.0%	0.0%
SF Express	Dalian	1	100.0%	100.0%	100.0%	100.0%	0.0%
SF Express	Tianjin	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Taiyuan	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Ningbo	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Guangzhou	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Chengdu	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Hangzhou	1	100.0%	0.0%	100.0%	100.0%	0.0%
SF Express	Guilin	1	100.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Wuhan	1	100.0%	0.0%	100.0%	100.0%	0.0%
SF Express	Cangzhou	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Jinan	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Haikou	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Shenzhen	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Weifang	1	100.0%	0.0%	0.0%	0.0%	100.0%
SF Express	Suzhou	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Jingzhou	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Yingkou	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Xi'an	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Xingtai	1	0.0%	0.0%	0.0%	0.0%	0.0%
SF Express	Changchun	1	0.0%	0.0%	0.0%	0.0%	0.0%

Role-level job identifiers

Table A6: Role-level job identifiers

No.	Brand	Platform	City	Location	Employer	Job title	Category	Employment
1	SF Express	BOSS Zhipin	Beijing	Shunyi District, Houshayu / Beijing	SF Express	SF Express Parcel Sorter	Express parcel sorting	SF direct / SF brand
2	SF Express	BOSS Zhipin	Hangzhou	Yuhang District, Renhe	SF Express	SF Express High-pay Parcel Sorter	Express parcel sorting	SF direct / SF brand
3	SF Express	BOSS Zhipin	Xiamen	Haicang District, Dongfu	SF Express	SF Express Parcel Sorter	Bulky-parcel sorting	SF direct / SF brand
4	SF Express	BOSS Zhipin	Sanya	Tianya District, Tianya Town	SF Express	SF Express Parcel Sorter	Express parcel sorting	SF direct / SF brand
5	SF Express	BOSS Zhipin	Changchun	Kuancheng District, Changchun Economic Development Zone	SF Express	SF Express Parcel Sorter	Trunk/branch-line and small-parcel sorting	SF direct / SF brand
6	SF Express	BOSS Zhipin	Jingzhou	Shashi District, Jingzhou Economic Development Zone	SF Express / SF-related listing	SF Express Part-time Sorting and Loading/Unloading Worker	Part-time sorting / loading-unloading	SF-related / pending verification
7	SF Express	BOSS Zhipin	Zhongshan	Sanjiao Town	SF Express	SF Express Direct-hire Parcel Sorter	Bulky-parcel sorting	SF direct / SF brand
8	SF Express	BOSS Zhipin	Jinan	Licheng District, Yaoqiang International Airport	Shenzhen Shunlu Logistics Co., Ltd.	SF Express Sorting and Loading/Unloading Worker	Sorting / loading-unloading / operations	SF-related / outsourced or platform labour
9	SF Express	BOSS Zhipin	Dongguan	Dongguan / Humen Town, Danning	Beijing Shuncheng Logistics Co. / Jinhua Jinfeng Express and others	SF Express Sorting/Forklift Worker	Sorting / forklift / transfer hub	SF-related / outsourced or platform labour
10	SF Express	BOSS Zhipin	Wuhan	Dongxihu District, Zouma Ling / Sandian	SF Express / Beijing Shuncheng Logistics related sample	SF Express Direct-hire Parcel Sorter	Express parcel sorting / transfer facility	SF direct / SF brand + platform sample
11	SF Express	BOSS Zhipin	Tianjin	Dongli District, Wanxin Subdistrict	SF Express	SF Express Direct-hire Parcel Sorter	Express sorting / scanning / loading-unloading	SF direct / SF brand
12	SF Express	BOSS Zhipin	Guilin	Guilin	Beijing Shuncheng Logistics Co. / SF-related	SF Express Sorting and Loading/Unloading Worker	Sorting / loading-unloading	SF-related / outsourced or platform labour
13	SF Express	BOSS Zhipin	Nanchang	Nanchang County, Xiangtang	Jiangxi SF Express Co., Ltd.	SF Express Temporary/short-term Sorting/Forklift Worker	Bulky-parcel sorting / forklift	SF direct / SF brand
14	SF Express	BOSS Zhipin	Xingtai	Xingtai / Nanhe Transfer Centre	SF Express / SF Nanhe Transfer Centre	SF Express Parcel Sorter	Express parcel sorting / hourly worker	SF direct / SF brand + related sample
15	SF Express	BOSS Zhipin	Hefei	Changfeng County, Beicheng / Economic Development Zone	SF Express / SF Express Co., Ltd.	SF Express Part-time Parcel Sorter	Part-time sorting / full-time sorting	SF direct / SF brand
16	SF Express	BOSS Zhipin	Weifang	Fangzi District, Fenghuang Subdistrict	SF Express	SF Express Part-time Parcel Sorter	Part-time sorting	SF direct / SF brand
17	SF Express	BOSS Zhipin	Dalian	Ganjingzi District, Zhoushuizi / Jinzhou District, Wanli	SF Express / Beijing Shuncheng Logistics related sample	SF Express Part-time Parcel Sorter	Sorting / part-time	SF direct / SF brand + related sample
18	SF Express	BOSS Zhipin	Ningbo	Haishu District, Duantang	SF Express	SF Express Parcel Sorter	Small-parcel transfer-hub sorting	SF direct / SF brand

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No.	Brand	Platform	City	Location	Employer	Job title	Category	Employment
19	SF Express	BOSS Zhipin	Xi'an	Weiyang District, Xi'an Economic Development Zone	SF Express	SF Express Parcel Sorter	Cold-chain warehouse sorting	SF direct / SF brand
20	SF Express	BOSS Zhipin	Taiyuan	Xiaodian District, Wucheng Road	SF Express	SF Express Parcel Sorter	Express parcel sorting	SF direct / SF brand
21	SF Express	BOSS Zhipin	Yingkou	Yingkou	SF Express	SF Express Parcel Sorter	Express sorting / loading-unloading	SF direct / SF brand
22	SF Express	BOSS Zhipin	Cangzhou	Cangzhou	Hebei SF Express Co., Ltd.	SF Express Parcel Sorter	Sorting / forklift / operations	SF direct / SF brand
23	SF Express	BOSS Zhipin	Haikou	Meilan District, Meilan Airport	SF Express	SF Express Parcel Sorter	Transfer-hub sorting	SF direct / SF brand
24	SF Express	BOSS Zhipin	Nanping	Jiayang	SF Express	SF Express Sorting and Loading/Unloading Worker	Loading-unloading / sorting	SF direct / SF brand
25	SF Express	BOSS Zhipin	Chengdu	Shuangliu District, Baijia	SF Express	SF Express Dispatch, Stock Handling, Picking and Shipping Worker	Warehouse picking/checking	SF direct / SF brand
26	SF Express	BOSS Zhipin	Nanjing	Qixia District, Xingang Development Zone	SF Freight	SF Express Parcel Sorter	Freight small-parcel sorting	SF Freight / SF-related
27	SF Express	BOSS Zhipin	Shenzhen	Longhua District, Guanlan	SF Express Co., Ltd.	SF Express Parcel Sorter	Logistics sorting	SF direct / SF brand
28	SF Express	BOSS Zhipin	Shanghai	Jinshan District, Jinshan	SF Express	SF Express Parcel Sorter	Pickup/delivery with sorting; excluded from main model	SF direct / SF brand
29	SF Express	BOSS Zhipin	Guangzhou	Baiyun / Haizhu etc.	Guangzhou SF Express / SF Express	SF Express Parcel Sorter	Not pure sorting / to be added	SF direct / SF brand
30	SF Express	BOSS Zhipin	Suzhou	Location not specified	No reliable SF Express sorting sample found	SF Express To be added	Coverage gap	To be added
1	JD Logistics	Liepin	Guangzhou	Huangpu District	JD Logistics	JD Logistics Direct-hire Regular Warehouse Operator	Full-time / regular	JD direct hire / regular employee
2	JD Logistics	BOSS Zhipin	Wuhan	Hanyang District, Zhoutou	JD Logistics	JD Logistics Sorting and Stock Handling Worker	Full-time / regular	Not direct hire / unclear
3	JD Logistics	BOSS Zhipin	Wuhan	Wuhan	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
4	JD Logistics	BOSS Zhipin	Wuhan	Wuhan	JD Logistics - Central China Region	JD Logistics Direct-hire Picking and Packing Warehouse Worker	Full-time / regular	JD direct hire / regular employee
5	JD Logistics	BOSS Zhipin	Wuhan	Wuhan	JD Logistics - Central China Region	JD Logistics Direct-hire Parcel Sorter	Full-time / regular	JD direct hire / regular employee
6	JD Logistics	BOSS Zhipin	Wuhan	Wuhan	JD Logistics	JD Logistics Parcel Sorter	Hourly / part-time	Not direct hire / unclear
7	JD Logistics	BOSS Zhipin	Guangzhou	Panyu District, Xinzao	JD Logistics - Central China Region	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
8	JD Logistics	BOSS Zhipin	Xingtai	Pingxiang County	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
9	JD Logistics	BOSS Zhipin	Cangzhou	Cangxian County, Bohai New Area	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
10	JD Logistics	BOSS Zhipin	Hefei	Changfeng County, Beicheng	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
11	JD Logistics	BOSS Zhipin	Jingzhou	Shashi District, Economic Development Zone	JD Logistics	JD Logistics Sorting and Loading/Unloading Worker	Full-time / regular	Not direct hire / unclear
12	JD Logistics	BOSS Zhipin	Wuhu	Wuhu	JD Logistics	JD Logistics Sorting and Loading/Unloading Worker	Full-time / regular	Not direct hire / unclear
13	JD Logistics	BOSS Zhipin	Tianjin	Binhai New Area, Haibin	JD Logistics	JD Logistics Dispatch, Stock Handling, Picking and Shipping Worker	Full-time / regular	Not direct hire / unclear
14	JD Logistics	BOSS Zhipin	Taiyuan	Xiaodian District	JD Logistics	JD Logistics Part-time Sorting and Loading/Unloading Worker	Hourly / part-time	Not direct hire / unclear
15	JD Logistics	BOSS Zhipin	Tianjin	Beichen District, Shuangjie	JD Logistics	JD Logistics Direct-hire Part-time High-pay Parcel Sorter	Hourly / part-time	JD direct hire / regular employee
16	JD Logistics	BOSS Zhipin	Quanzhou	Jinjiang City, Guanqiao Town	JD Logistics	JD Logistics Regular Parcel Sorter	Full-time / regular	JD direct hire / regular employee
17	JD Logistics	BOSS Zhipin	Zibo	Zibo	JD Logistics	JD Logistics Direct-hire Parcel Sorter	Full-time / regular	JD direct hire / regular employee
18	JD Logistics	BOSS Zhipin	Guangzhou	Huangpu District, Nangang	JD Logistics	JD Logistics Temporary/short-term Picking and Packing Warehouse Worker	Temporary worker	Not direct hire / unclear
19	JD Logistics	BOSS Zhipin	Wenzhou	Rui'an City, Xianjiang	JD Logistics	JD Logistics Direct-hire Part-time Parcel Sorter	Hourly / part-time	JD direct hire / regular employee
20	JD Logistics	BOSS Zhipin	Wuxi	Xishan District, Donggang	JD Logistics	JD Logistics Direct-hire Sorting and Loading/Unloading Worker	Full-time / regular	JD direct hire / regular employee
21	JD Logistics	BOSS Zhipin	Suzhou	Kunshan City, Qiangdeng	JD Logistics	JD Logistics Regular Sorting and Loading/Unloading Worker	Full-time / regular / part-time	JD direct hire / regular employee
22	JD Logistics	BOSS Zhipin	Xuzhou	Gulou District, Xianghe	JD Logistics	JD Logistics Direct-hire Part-time Parcel Sorter	Hourly / part-time	JD direct hire / regular employee
23	JD Logistics	BOSS Zhipin	Shanghai	Qingpu District, Baihe	JD Logistics	JD Logistics Sorting and Stock Handling Worker	Full-time / regular	JD direct hire / regular employee
24	JD Logistics	BOSS Zhipin	Dalian	Dalian	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
25	JD Logistics	BOSS Zhipin	Zhengzhou	Zhengzhou	JD Logistics	JD Logistics Direct-hire Regular Sorting and Loading/Unloading Worker	Full-time / regular	JD direct hire / regular employee
26	JD Logistics	BOSS Zhipin	Shanghai	Songjiang District, Shihudang	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
27	JD Logistics	BOSS Zhipin	Hefei	Changfeng County, Beicheng	JD Logistics	JD Logistics Direct-hire Part-time Parcel Sorter	Hourly / part-time	JD direct hire / regular employee
28	JD Logistics	BOSS Zhipin	Chengdu	Jinjiang District, Niushikou	JD Logistics	JD Logistics Regular Store/Front-warehouse Picking and Packing Worker	Full-time / regular	JD direct hire / regular employee
29	JD Logistics	BOSS Zhipin	Guiyang	Guanshanhu District, Southwest Trade City	JD Logistics	JD Logistics Direct-hire Sorting and Loading/Unloading Worker	Full-time / regular	JD direct hire / regular employee
30	JD Logistics	BOSS Zhipin	Wuxi	Xishan District, Donggang	JD Logistics	JD Logistics Regular Parcel Sorter	Full-time / regular	JD direct hire / regular employee
31	JD Logistics	BOSS Zhipin	Ningbo	Ningbo	JD Express	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
32	JD Logistics	BOSS Zhipin	Yanbian	Yanji City, Henan Subdistrict	JD Express	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
33	JD Logistics	BOSS Zhipin	Ningbo	Ningbo	JD Express	JD Logistics Sorting and Loading/Unloading Worker	Full-time / regular	Not direct hire / unclear
34	JD Logistics	BOSS Zhipin	Zhongshan	Dongsheng Town	JD Express	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear

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No.	Brand	Platform	City	Location	Employer	Job title	Category	Employment
35	JD Logistics	BOSS Zhipin	Nantong	Haimen District	JD Express	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
36	JD Logistics	BOSS Zhipin	Fuzhou	Fuzhou	JD Express	JD Logistics Sorting and Stock Handling Worker	Full-time / regular	Not direct hire / unclear
37	JD Logistics	BOSS Zhipin	Chengdu	Chengdu	JD Logistics	JD Logistics Part-time Parcel Sorter	Hourly / part-time	Not direct hire / unclear
38	JD Logistics	BOSS Zhipin	Nanchang	Nanchang County, Xiangtang	JD Logistics	JD Logistics Temporary/short-term Sorting and Stock Handling Worker	Short-term / temporary	Not direct hire / unclear
39	JD Logistics	BOSS Zhipin	Tianjin	Wuqing District	JD Logistics	JD Logistics Part-time Parcel Sorter	Hourly / part-time	Not direct hire / unclear
40	JD Logistics	BOSS Zhipin	Taizhou	Hailing District	JD Logistics	JD Logistics Dispatch, Stock Handling, Picking and Shipping Worker	Hourly / part-time	Not direct hire / unclear
41	JD Logistics	BOSS Zhipin	Suzhou	Kunshan City, Yushan	JD Logistics	JD Logistics Temporary/short-term Regular Picking and Packing Warehouse Worker	Full-time / regular / temporary	Not direct hire / unclear
42	JD Logistics	BOSS Zhipin	Guangzhou	Panyu District, Xinzao	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
43	JD Logistics	BOSS Zhipin	Urumqi	Toutunhe District	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
44	JD Logistics	BOSS Zhipin	Wuxi	Xishan District, Donggang	JD Logistics	JD Logistics Picking and Packing Warehouse Worker	Full-time / part-time	Not direct hire / unclear
45	JD Logistics	BOSS Zhipin	Luoyang	Luoyang	JD Logistics	JD Logistics Direct-hire Parcel Sorter	Full-time / regular	JD direct hire / regular employee
46	JD Logistics	BOSS Zhipin	Guangzhou	Zengcheng District, Xintang	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
47	JD Logistics	BOSS Zhipin	Chengdu	Qingbaijiang District, Hongyang	JD Logistics	JD Logistics Direct-hire Parcel Sorter	Full-time / regular	JD direct hire / regular employee
48	JD Logistics	BOSS Zhipin	Nanning	Nanning	Jingbangda Trading Co., Ltd.	JD Logistics Picking and Packing Warehouse Worker	Hourly / part-time	Not direct hire / unclear
49	JD Logistics	BOSS Zhipin	Guangzhou	Zengcheng District, Xintang	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	JD direct hire / regular employee
50	JD Logistics	BOSS Zhipin	Zhuhai	Xiangzhou District, Hengqin	JD Logistics	JD Logistics Direct-hire Regular Warehouse Worker	Full-time / regular	JD direct hire / regular employee
51	JD Logistics	BOSS Zhipin	Chengdu	Xindu District, Banzhuyuan	JD Logistics	JD Logistics Direct-hire Warehouse Worker	Full-time / regular	JD direct hire / regular employee
52	JD Logistics	BOSS Zhipin	Chengdu	Shuangliu District, Xinxing	JD Logistics	JD Logistics Warehouse Operator	Full-time / regular	JD direct hire / regular employee
53	JD Logistics	BOSS Zhipin	Chengdu	Shuangliu District, Baijia	JD Logistics	JD Logistics Warehouse Worker	Full-time / regular	Not direct hire / unclear
54	JD Logistics	BOSS Zhipin	Hangzhou	Fuyang District, Dongzhou	JD Logistics	JD Logistics Direct-hire Regular Parcel Sorter	Full-time / regular	JD direct hire / regular employee
55	JD Logistics	BOSS Zhipin	Beijing	Shunyi District, Shiyuan	JD Logistics	JD Logistics Direct-hire Temporary/short-term Parcel Sorter	Temporary worker	JD direct hire / regular employee
56	JD Logistics	BOSS Zhipin	Beijing	Fengtai District, Lugouqiao	JD Logistics	JD Logistics Store/Front-warehouse Picking and Packing Worker	Full-time / regular	JD direct hire / regular employee
57	JD Logistics	BOSS Zhipin	Beijing	Fengtai District, Lugouqiao	JD Logistics	JD Logistics Direct-hire Store/Front-warehouse Picking and Packing Worker	Full-time / regular	JD direct hire / regular employee
58	JD Logistics	BOSS Zhipin	Shanghai	Pudong New Area, Chuansha	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
59	JD Logistics	BOSS Zhipin	Shanghai	Pudong New Area, Chuansha	JD Logistics	JD Logistics Part-time Parcel Sorter	Part-time	Not direct hire / unclear
60	JD Logistics	BOSS Zhipin	Shanghai	Songjiang District, Shihudang	JD Logistics	JD Logistics Direct-hire Parcel Sorter	Full-time / regular	JD direct hire / regular employee
61	JD Logistics	BOSS Zhipin	Nanjing	Lishui District, Zhetang	JD Logistics	JD Logistics High-pay Warehouse Operator	Full-time / regular	Not direct hire / unclear
62	JD Logistics	BOSS Zhipin	Zhenjiang	Zhenjiang	JD Logistics	JD Logistics Regular Warehouse Worker	Full-time / regular	Not direct hire / unclear
63	JD Logistics	BOSS Zhipin	Nanning	Xingning District, Wutang	JD Logistics	JD Logistics Warehouse Worker	Full-time / regular	Not direct hire / unclear
64	JD Logistics	BOSS Zhipin	Hefei	Changfeng County, Beicheng	JD Logistics	JD Logistics Direct-hire Warehouse Worker	Full-time / regular	JD direct hire / regular employee
65	JD Logistics	BOSS Zhipin	Nanchang	Nanchang County, Xiangtang	Jingbangda Trading Co., Ltd.	JD Logistics Direct-hire Warehouse Operator	Full-time / regular	JD direct hire / regular employee
66	JD Logistics	BOSS Zhipin	Chongqing	Banan District, Jieshi	JD Logistics	JD Logistics Dispatch, Stock Handling, Picking and Shipping Worker	Full-time / regular	Not direct hire / unclear
67	JD Logistics	BOSS Zhipin	Zhuhai	Xiangzhou District, Hengqin	JD Logistics	JD Logistics Parcel Sorter	Full-time / regular	Not direct hire / unclear
68	JD Logistics	BOSS Zhipin	Shenyang	Shenyang	JD Logistics	JD Logistics Warehouse Operator	Full-time / regular	JD direct hire / regular employee

Role-level salary levels

Table A7: Role-level salary levels

No.	Brand	City	Salary range	Basis	Months	Monthly low	Monthly high	Monthly midpoint	Annual cash wage	Employer cost
1	SF Express	Beijing	RMB 7,000--8,000/month	Monthly salary		RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
2	SF Express	Hangzhou	RMB 6,000--8,000/month	Monthly salary		RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
3	SF Express	Xiamen	RMB 7,000--8,000/month	Monthly salary		RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
4	SF Express	Sanya	RMB 5,000--6,000/month	Monthly salary		RMB 5,000	RMB 6,000	RMB 5,500	RMB 66,000	RMB 85,800
5	SF Express	Changchun	RMB 6,000--7,000/month	Monthly salary		RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
6	SF Express	Jingzhou	RMB 10--20/hour; monthly equivalent RMB 2,600--5,200	Hourly wage		RMB 2,600	RMB 5,200	RMB 3,900	RMB 46,800	RMB 60,840
7	SF Express	Zhongshan	RMB 6,000--8,000/month	Monthly salary		RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
8	SF Express	Jinan	RMB 5,000--8,000/month	Monthly salary		RMB 5,000	RMB 8,000	RMB 6,500	RMB 78,000	RMB 101,400
9	SF Express	Dongguan	RMB 7,000--8,000/month	Monthly salary		RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
10	SF Express	Wuhan	RMB 6,000--7,000/month	Monthly salary		RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400

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No.	Brand	City	Salary range	Basis	Months	Monthly low	Monthly high	Monthly midpoint	Annual cash wage	Employer cost
11	SF Express	Tianjin	RMB 6,000--8,000/ month	Monthly salary		RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
12	SF Express	Guilin	RMB 4,000--7,000/ month	Monthly salary		RMB 4,000	RMB 7,000	RMB 5,500	RMB 66,000	RMB 85,800
13	SF Express	Nanchang	RMB 200--400/day; monthly equivalent RMB 5,200--10,400	Daily wage		RMB 5,200	RMB 10,400	RMB 7,800	RMB 93,600	RMB 121,680
14	SF Express	Xingtai	RMB 5,000--7,000/ month	Monthly salary		RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
15	SF Express	Hefei	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage		RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
16	SF Express	Weifang	RMB 20--25/hour; monthly equivalent RMB 2,600--3,250	Hourly wage		RMB 2,600	RMB 3,250	RMB 2,925	RMB 35,100	RMB 45,630
17	SF Express	Dalian	RMB 5,000--7,000/ month	Monthly salary		RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
18	SF Express	Ningbo	RMB 7,000--9,000/ month	Monthly salary		RMB 7,000	RMB 9,000	RMB 8,000	RMB 96,000	RMB 124,800
19	SF Express	Xi'an	RMB 6,000--8,000/ month	Monthly salary		RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
20	SF Express	Taiyuan	RMB 5,000--7,000/ month	Monthly salary		RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
21	SF Express	Yingkou	RMB 4,000--6,000/ month	Monthly salary		RMB 4,000	RMB 6,000	RMB 5,000	RMB 60,000	RMB 78,000
22	SF Express	Cangzhou	RMB 5,000--6,000/ month	Monthly salary		RMB 5,000	RMB 6,000	RMB 5,500	RMB 66,000	RMB 85,800
23	SF Express	Haikou	RMB 5,000--7,000/ month	Monthly salary		RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
24	SF Express	Nanping	RMB 6,000--7,000/ month	Monthly salary		RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
25	SF Express	Chengdu	RMB 6,000--8,000/ month	Monthly salary		RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
26	SF Express	Nanjing	RMB 7,000--8,000/ month	Monthly salary		RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
27	SF Express	Shenzhen	RMB 7,000--8,000/ month	Monthly salary		RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
28	SF Express	Shanghai	RMB 6,000--9,000/ month	Monthly salary		RMB 6,000	RMB 9,000	RMB 7,500	RMB 90,000	RMB 117,000
29	SF Express	Guangzhou	RMB 5,000--8,000/ month	Monthly salary		RMB 5,000	RMB 8,000	RMB 6,500	RMB 78,000	RMB 101,400
30	SF Express	Suzhou	To be added	To be added						
1	JD Logistics	Guangzhou	RMB 6,000--9,000/ month $\times$ 13 months	Monthly salary	13	RMB 6,000	RMB 9,000	RMB 7,500	RMB 97,500	RMB 126,750
2	JD Logistics	Wuhan	RMB 6,000--10,000/ month	Monthly salary	12	RMB 6,000	RMB 10,000	RMB 8,000	RMB 96,000	RMB 124,800
3	JD Logistics	Wuhan	RMB 6,000--7,000/ month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
4	JD Logistics	Wuhan	RMB 5,000--7,000/ month	Monthly salary	12	RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
5	JD Logistics	Wuhan	RMB 5,000--8,000/ month	Monthly salary	12	RMB 5,000	RMB 8,000	RMB 6,500	RMB 78,000	RMB 101,400
6	JD Logistics	Wuhan	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
7	JD Logistics	Guangzhou	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
8	JD Logistics	Xingtai	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
9	JD Logistics	Cangzhou	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
10	JD Logistics	Hefei	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
11	JD Logistics	Jingzhou	RMB 6,000--7,000/ month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
12	JD Logistics	Wuhu	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
13	JD Logistics	Tianjin	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
14	JD Logistics	Taiyuan	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
15	JD Logistics	Tianjin	RMB 25--30/hour; monthly equivalent RMB 6,500--7,800	Hourly wage	12	RMB 6,500	RMB 7,800	RMB 7,150	RMB 85,800	RMB 111,540
16	JD Logistics	Quanzhou	RMB 7,000--10,000/ month	Monthly salary	12	RMB 7,000	RMB 10,000	RMB 8,500	RMB 102,000	RMB 132,600
17	JD Logistics	Zibo	RMB 6,000--7,000/ month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
18	JD Logistics	Guangzhou	RMB 6,000--7,000/ month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
19	JD Logistics	Wenzhou	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
20	JD Logistics	Wuxi	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
21	JD Logistics	Suzhou	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
22	JD Logistics	Xuzhou	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
23	JD Logistics	Shanghai	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
24	JD Logistics	Dalian	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
25	JD Logistics	Zhengzhou	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
26	JD Logistics	Shanghai	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
27	JD Logistics	Hefei	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
28	JD Logistics	Chengdu	RMB 5,000--6,000/ month	Monthly salary	12	RMB 5,000	RMB 6,000	RMB 5,500	RMB 66,000	RMB 85,800
29	JD Logistics	Guiyang	RMB 5,000--9,000/ month	Monthly salary	12	RMB 5,000	RMB 9,000	RMB 7,000	RMB 84,000	RMB 109,200
30	JD Logistics	Wuxi	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
31	JD Logistics	Ningbo	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
32	JD Logistics	Yanbian	RMB 4,000--5,000/ month	Monthly salary	12	RMB 4,000	RMB 5,000	RMB 4,500	RMB 54,000	RMB 70,200
33	JD Logistics	Ningbo	RMB 6,000--8,000/ month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
34	JD Logistics	Zhongshan	RMB 6,000--7,000/ month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
35	JD Logistics	Nantong	RMB 7,000--8,000/ month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000

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No.	Brand	City	Salary range	Basis	Months	Monthly low	Monthly high	Monthly midpoint	Annual cash wage	Employer cost
36	JD Logistics	Fuzhou	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
37	JD Logistics	Chengdu	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
38	JD Logistics	Nanchang	RMB 5,000--7,000/month	Monthly salary	12	RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
39	JD Logistics	Tianjin	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
40	JD Logistics	Taizhou	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
41	JD Logistics	Suzhou	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
42	JD Logistics	Guangzhou	RMB 7,000--8,000/month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
43	JD Logistics	Urumqi	RMB 5,000--7,000/month	Monthly salary	12	RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
44	JD Logistics	Wuxi	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
45	JD Logistics	Luoyang	RMB 5,000--6,000/month	Monthly salary	12	RMB 5,000	RMB 6,000	RMB 5,500	RMB 66,000	RMB 85,800
46	JD Logistics	Guangzhou	RMB 6,000--10,000/month	Monthly salary	12	RMB 6,000	RMB 10,000	RMB 8,000	RMB 96,000	RMB 124,800
47	JD Logistics	Chengdu	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
48	JD Logistics	Nanning	RMB 20--25/hour; monthly equivalent RMB 5,200--6,500	Hourly wage	12	RMB 5,200	RMB 6,500	RMB 5,850	RMB 70,200	RMB 91,260
49	JD Logistics	Guangzhou	RMB 7,000--10,000/month	Monthly salary	12	RMB 7,000	RMB 10,000	RMB 8,500	RMB 102,000	RMB 132,600
50	JD Logistics	Zhuhai	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
51	JD Logistics	Chengdu	RMB 5,000--7,000/month	Monthly salary	12	RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
52	JD Logistics	Chengdu	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
53	JD Logistics	Chengdu	RMB 5,000--8,000/month	Monthly salary	12	RMB 5,000	RMB 8,000	RMB 6,500	RMB 78,000	RMB 101,400
54	JD Logistics	Hangzhou	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
55	JD Logistics	Beijing	RMB 4,000--5,000/month	Monthly salary	12	RMB 4,000	RMB 5,000	RMB 4,500	RMB 54,000	RMB 70,200
56	JD Logistics	Beijing	RMB 7,000--8,000/month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
57	JD Logistics	Beijing	RMB 6,000--7,000/month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
58	JD Logistics	Shanghai	RMB 7,000--8,000/month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
59	JD Logistics	Shanghai	RMB 6,000--7,000/month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
60	JD Logistics	Shanghai	RMB 7,000--8,000/month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
61	JD Logistics	Nanjing	RMB 7,000--8,000/month	Monthly salary	12	RMB 7,000	RMB 8,000	RMB 7,500	RMB 90,000	RMB 117,000
62	JD Logistics	Zhenjiang	RMB 6,000--9,000/month	Monthly salary	12	RMB 6,000	RMB 9,000	RMB 7,500	RMB 90,000	RMB 117,000
63	JD Logistics	Nanning	RMB 5,000--7,000/month	Monthly salary	12	RMB 5,000	RMB 7,000	RMB 6,000	RMB 72,000	RMB 93,600
64	JD Logistics	Hefei	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
65	JD Logistics	Nanchang	RMB 6,000--7,000/month	Monthly salary	12	RMB 6,000	RMB 7,000	RMB 6,500	RMB 78,000	RMB 101,400
66	JD Logistics	Chongqing	RMB 5,000--6,000/month	Monthly salary	12	RMB 5,000	RMB 6,000	RMB 5,500	RMB 66,000	RMB 85,800
67	JD Logistics	Zhuhai	RMB 6,000--8,000/month	Monthly salary	12	RMB 6,000	RMB 8,000	RMB 7,000	RMB 84,000	RMB 109,200
68	JD Logistics	Shenyang	RMB 8,000--10,000/month	Monthly salary	12	RMB 8,000	RMB 10,000	RMB 9,000	RMB 108,000	RMB 140,400

Role-level salary components and benefits

Table A8: Role-level salary components and benefits

No.	Brand	City	Night	Direct	Insurance	Fund	Housing	Meals	Overtime	Piece/perf.	Bonus/13th	Listed components
1	SF Express	Beijing	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
2	SF Express	Hangzhou	Yes	Yes	No	No	Yes	Yes	No	No	No	Night shift / shift rotation; Meal allowance / meals provided; Accommodation provided; Direct hire / regular employment
3	SF Express	Xiamen	Yes	Yes	Yes	No	Yes	No	No	Yes	No	Night shift / shift rotation; Accommodation provided; Social insurance; Piece-rate / performance pay; Direct hire / regular employment
4	SF Express	Sanya	Yes	Yes	No	No	No	No	Yes	No	No	Night shift / shift rotation; Overtime; Direct hire / regular employment
5	SF Express	Changchun	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
6	SF Express	Jingzhou	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
7	SF Express	Zhongshan	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
8	SF Express	Jinan	No	No	No	No	No	No	No	No	No	Not disclosed
9	SF Express	Dongguan	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
10	SF Express	Wuhan	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Night shift / shift rotation; Meal allowance / meals provided; Accommodation provided; Bonus / allowance / 13th-month pay; Direct hire / regular employment
11	SF Express	Tianjin	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
12	SF Express	Guilin	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
13	SF Express	Nanchang	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
14	SF Express	Xingtai	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment

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No.	Brand	City	Night	Direct	Insurance	Fund	Housing	Meals	Overtime	Piece/ perf.	Bonus/ 13th	Listed components
15	SF Ex- press	Hefei	Yes	Yes	Yes	No	Yes	Yes	No	No	Yes	Night shift / shift rotation; Meal allowance / meals provided; Accommodation provided; Social insurance; Bonus / allowance / 13th-month pay; Direct hire / regular employment
16	SF Ex- press	Weifang	Yes	Yes	No	No	No	No	No	Yes	No	Night shift / shift rotation; Piece-rate / performance pay; Direct hire / regular employment
17	SF Ex- press	Dalian	Yes	Yes	Yes	No	Yes	Yes	No	No	No	Night shift / shift rotation; Meal allowance / meals provided; Accommodation provided; Social insurance; Direct hire / regular employment
18	SF Ex- press	Ningbo	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
19	SF Ex- press	Xi'an	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
20	SF Ex- press	Taiyuan	Yes	Yes	No	No	No	No	Yes	No	No	Night shift / shift rotation; Overtime; Direct hire / regular employment
21	SF Ex- press	Yingkou	No	Yes	No	No	No	No	No	No	Yes	Bonus / allowance / 13th-month pay; Direct hire / regular employment
22	SF Ex- press	Cangzhou	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
23	SF Ex- press	Haikou	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
24	SF Ex- press	Nanping	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
25	SF Ex- press	Chengdu	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
26	SF Ex- press	Nanjing	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
27	SF Ex- press	Shenzhen	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
28	SF Ex- press	Shanghai	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
29	SF Ex- press	Guangzhou	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
30	SF Ex- press	Suzhou	No	No	No	No	No	No	No	No	No	Not disclosed
1	JD Logis- tics	Guangzhou	No	Yes	Yes	Yes	No	No	No	No	Yes	Social insurance; Housing fund; Bonus / allowance / 13th-month pay; Direct hire / regular employment
2	JD Logis- tics	Wuhan	No	No	Yes	No	No	Yes	Yes	No	No	Meal allowance / meals provided; Social insurance; Overtime
3	JD Logis- tics	Wuhan	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
4	JD Logis- tics	Wuhan	No	Yes	Yes	Yes	No	No	No	No	No	Social insurance; Housing fund; Direct hire / regular employment
5	JD Logis- tics	Wuhan	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
6	JD Logis- tics	Wuhan	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
7	JD Logis- tics	Guangzhou	No	Yes	Yes	Yes	No	No	No	No	Yes	Social insurance; Housing fund; Bonus / allowance / 13th-month pay; Direct hire / regular employment
8	JD Logis- tics	Xingtai	Yes	No	No	No	Yes	Yes	No	Yes	Yes	Night shift / shift rotation; Meal allowance / meals provided; Accommodation provided; Piece-rate / performance pay; Bonus / allowance / 13th-month pay
9	JD Logis- tics	Cangzhou	No	Yes	Yes	Yes	No	Yes	No	No	Yes	Meal allowance / meals provided; Social insurance; Housing fund; Bonus / allowance / 13th-month pay; Direct hire / regular employment
10	JD Logis- tics	Hefei	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
11	JD Logis- tics	Jingzhou	Yes	No	Yes	Yes	No	No	No	No	No	Night shift / shift rotation; Social insurance; Housing fund
12	JD Logis- tics	Wuhu	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
13	JD Logis- tics	Tianjin	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
14	JD Logis- tics	Taiyuan	Yes	No	No	No	No	No	No	No	Yes	Night shift / shift rotation; Bonus / allowance / 13th-month pay
15	JD Logis- tics	Tianjin	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
16	JD Logis- tics	Quanzhou	No	Yes	Yes	Yes	Yes	No	No	No	No	Accommodation provided; Social insurance; Housing fund; Direct hire / regular employment
17	JD Logis- tics	Zibo	Yes	Yes	Yes	Yes	No	No	No	No	No	Night shift / shift rotation; Social insurance; Housing fund; Direct hire / regular employment
18	JD Logis- tics	Guangzhou	No	No	No	No	No	Yes	Yes	No	No	Meal allowance / meals provided; Overtime
19	JD Logis- tics	Wenzhou	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
20	JD Logis- tics	Wuxi	No	Yes	No	No	No	No	No	Yes	Yes	Piece-rate / performance pay; Bonus / allowance / 13th-month pay; Direct hire / regular employment
21	JD Logis- tics	Suzhou	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
22	JD Logis- tics	Xuzhou	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
23	JD Logis- tics	Shanghai	No	Yes	Yes	Yes	No	No	No	No	No	Social insurance; Housing fund; Direct hire / regular employment
24	JD Logis- tics	Dalian	Yes	No	Yes	Yes	No	No	No	No	No	Night shift / shift rotation; Social insurance; Housing fund
25	JD Logis- tics	Zhengzhou	Yes	Yes	Yes	Yes	No	No	No	No	No	Night shift / shift rotation; Social insurance; Housing fund; Direct hire / regular employment
26	JD Logis- tics	Shanghai	No	Yes	Yes	No	Yes	No	No	No	No	Accommodation provided; Social insurance; Direct hire / regular employment
27	JD Logis- tics	Hefei	Yes	Yes	No	No	No	No	No	No	Yes	Night shift / shift rotation; Bonus / allowance / 13th-month pay; Direct hire / regular employment
28	JD Logis- tics	Chengdu	No	Yes	Yes	Yes	No	No	No	No	No	Social insurance; Housing fund; Direct hire / regular employment
29	JD Logis- tics	Guiyang	No	Yes	Yes	Yes	No	No	No	No	No	Social insurance; Housing fund; Direct hire / regular employment

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No.	Brand	City	Night	Direct	Insurance	Fund	Housing	Meals	Overtime	Piece/ perf.	Bonus/ 13th	Listed components
30	JD Logis- tics	Wuxi	No	Yes	No	No	No	Yes	No	Yes	Yes	Meal allowance / meals provided; Piece-rate / performance pay; Bonus / allowance / 13th-month pay; Direct hire / regular employment
31	JD Logis- tics	Ningbo	No	No	No	No	No	No	No	No	No	Not disclosed
32	JD Logis- tics	Yanbian	No	No	No	No	No	No	No	No	No	Not disclosed
33	JD Logis- tics	Ningbo	No	No	No	No	No	No	No	No	No	Not disclosed
34	JD Logis- tics	Zhongshan	No	No	No	No	No	No	No	No	No	Not disclosed
35	JD Logis- tics	Nantong	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
36	JD Logis- tics	Fuzhou	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
37	JD Logis- tics	Chengdu	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
38	JD Logis- tics	Nanchang	No	No	No	No	No	No	No	No	No	Not disclosed
39	JD Logis- tics	Tianjin	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
40	JD Logis- tics	Taizhou	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
41	JD Logis- tics	Suzhou	No	No	Yes	Yes	No	No	No	Yes	No	Social insurance; Housing fund; Piece-rate / performance pay
42	JD Logis- tics	Guangzhou	No	No	No	No	No	No	No	No	No	Not disclosed
43	JD Logis- tics	Urumqi	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
44	JD Logis- tics	Wuxi	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
45	JD Logis- tics	Luoyang	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
46	JD Logis- tics	Guangzhou	No	Yes	Yes	Yes	Yes	Yes	No	No	No	Meal allowance / meals provided; Accommodation provided; Social insurance; Housing fund; Direct hire / regular employment
47	JD Logis- tics	Chengdu	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
48	JD Logis- tics	Nanning	Yes	No	No	No	No	No	No	No	Yes	Night shift / shift rotation; Bonus / allowance / 13th-month pay
49	JD Logis- tics	Guangzhou	No	Yes	Yes	Yes	No	No	No	No	Yes	Social insurance; Housing fund; Bonus / allowance / 13th-month pay; Direct hire / regular employment
50	JD Logis- tics	Zhuhai	No	Yes	Yes	Yes	Yes	No	No	No	No	Accommodation provided; Social insurance; Housing fund; Direct hire / regular employment
51	JD Logis- tics	Chengdu	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
52	JD Logis- tics	Chengdu	No	Yes	Yes	Yes	No	No	No	No	No	Social insurance; Housing fund; Direct hire / regular employment
53	JD Logis- tics	Chengdu	No	No	No	No	No	No	No	No	No	Not disclosed
54	JD Logis- tics	Hangzhou	No	Yes	Yes	Yes	Yes	Yes	No	No	No	Meal allowance / meals provided; Accommodation provided; Social insurance; Housing fund; Direct hire / regular employment
55	JD Logis- tics	Beijing	Yes	Yes	No	No	No	No	No	No	No	Night shift / shift rotation; Direct hire / regular employment
56	JD Logis- tics	Beijing	No	Yes	No	No	No	No	No	No	No	Direct hire / regular employment
57	JD Logis- tics	Beijing	No	Yes	Yes	Yes	No	No	No	No	No	Social insurance; Housing fund; Direct hire / regular employment
58	JD Logis- tics	Shanghai	Yes	No	No	No	No	No	No	No	No	Night shift / shift rotation
59	JD Logis- tics	Shanghai	No	No	No	No	No	No	No	No	No	Not disclosed
60	JD Logis- tics	Shanghai	No	Yes	Yes	Yes	Yes	No	No	No	No	Accommodation provided; Social insurance; Housing fund; Direct hire / regular employment
61	JD Logis- tics	Nanjing	No	No	No	No	No	No	No	Yes	Yes	Piece-rate / performance pay; Bonus / allowance / 13th-month pay
62	JD Logis- tics	Zhenjiang	No	No	Yes	No	No	No	No	No	No	Social insurance
63	JD Logis- tics	Nanning	No	No	No	No	No	No	No	No	No	Not disclosed
64	JD Logis- tics	Hefei	No	Yes	No	No	No	No	Yes	No	No	Overtime; Direct hire / regular employment
65	JD Logis- tics	Nanchang	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Night shift / shift rotation; Meal allowance / meals provided; Accommodation provided; Social insurance; Housing fund; Direct hire / regular employment
66	JD Logis- tics	Chongqing	No	No	No	No	No	No	No	No	No	Not disclosed
67	JD Logis- tics	Zhuhai	No	No	No	No	Yes	No	No	No	No	Accommodation provided
68	JD Logis- tics	Shenyang	No	Yes	No	No	Yes	No	No	No	No	Accommodation provided; Direct hire / regular employment

Role-level shift and work-content information

Table A9: Role-level shift and work-content information

No.	Brand	City	Shift summary	Hours/ day	Days/ month	Work content	Clean notes
1	SF Ex- press	Beijing	Night or rotating shift reported			Sorting; Scanning; Platform hourly worker; Able to work night shifts / Transfer hub	Grade A sample
2	SF Ex- press	Hangzhou	Night or rotating shift reported; time window(s): 20:30-08:30; 12 hours/day reported	12.0		Sorting; Bagging/bundling; Automation-assisted; Evening/night shift; Meal allowance; Accommodation	Grade A sample
3	SF Ex- press	Xiamen	Night or rotating shift reported; time window(s): 20:00-06:30; 10.5 hours/day reported	10.5		Sorting; Loading/unloading; Night shift; Accommodation provided; Five social insurances; Commission/accrual pay	Grade A sample

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No.	Brand	City	Shift summary	Hours/ day	Days/ month	Work content	Clean notes
4	SF Ex-press	Sanya	Night or rotating shift reported			Sorting; Rotating shift; Overtime pay	Grade A sample
5	SF Ex-press	Changchun	No night/rotating shift disclosed			Sorting; Scanning; Stacking; Unbagging; Bag dumping; Bag building	Grade A sample
6	SF Ex-press	Jingzhou	No night/rotating shift disclosed			Sorting; Loading/unloading; Part-time; Transfer hub; Scanning	Grade B sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity; Monthly equivalent was converted from hourly/daily wage assumptions
7	SF Ex-press	Zhongshan	Night or rotating shift reported; time window(s): 20:00-07:00; 11 hours/day reported; 26 days/month reported	11.0	26.0	Sorting; Bulky parcels; Night shift; 4 rest days/month	Grade A sample
8	SF Ex-press	Jinan	No night/rotating shift disclosed			Sorting; Loading/unloading; Operations; Airport; SOP; Small parcels	Grade B sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
9	SF Ex-press	Dongguan	Night or rotating shift reported			Sorting; Forklift; Night shift; Transfer hub	Grade B sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
10	SF Ex-press	Wuhan	Night or rotating shift reported; time window(s): 4-6; 12 hours/day reported; 26 days/month reported	12.0	26.0	Sorting; Scanning; Bag building; Two-shift rotation; Meals and accommodation provided; Meal allowance; Night-shift allowance	Grade A sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
11	SF Ex-press	Tianjin	No night/rotating shift disclosed			Sorting; Scanning; Loading/unloading; Bag building/unbuilding; Cage/trolley operation	Grade A sample
12	SF Ex-press	Guilin	Night or rotating shift reported			Sorting; Loading/unloading; Two-shift rotation; Transfer hub / Distribution point	Grade A sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
13	SF Ex-press	Nanchang	Night or rotating shift reported; 26 days/month reported		26.0	Sorting; Forklift; Short-term worker; Daily wage; Night shift	Grade A sample; Monthly equivalent was converted from hourly/daily wage assumptions
14	SF Ex-press	Xingtai	No night/rotating shift disclosed; time window(s): 12:00-24:00; 12 hours/day reported	12.0		Sorting; Small-parcel bag building; Hourly worker; Transfer centre	Grade A sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
15	SF Ex-press	Hefei	Night or rotating shift reported			Sorting; Stacking parcels; Plastic wrapping; Parcel feeding; Hourly worker; Night-shift allowance; Meal allowance; Accommodation provided; Five social insurances	Grade A sample; Monthly equivalent was converted from hourly/daily wage assumptions
16	SF Ex-press	Weifang	Night or rotating shift reported; time window(s): 19:00-24:00; 5 hours/day reported	5.0		Sorting; Part-time; Bulky parcels; Quantity-based pay; Half-month settlement; Night shift	Grade A sample; Monthly equivalent was converted from hourly/daily wage assumptions
17	SF Ex-press	Dalian	Night or rotating shift reported; time window(s): 15:30-01:30; 10 hours/day reported; 26 days/month reported	10.0	26.0	Sorting; Scanning; Evening/night shift; Five social insurances; Accommodation; Meal subsidy; 4-6 rest days/month	Grade A sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
18	SF Ex-press	Ningbo	Night or rotating shift reported; time window(s): 01:00-13:30; 12.5 hours/day reported	12.5		Sorting; Small parcels; Transfer hub; Night shift / Early morning; Weekly settlement; RMB 22 / hour	Grade A sample
19	SF Ex-press	Xi'an	No night/rotating shift disclosed			Cold-chain; Picking; Put-to-order sorting; Checking/rechecking; Packing; Scanning	Grade A sample
20	SF Ex-press	Taiyuan	Night or rotating shift reported; time window(s): 8-12, 4-6; 10 hours/day reported; 25 days/month reported	10.0	25.0	Sorting; Rotating shift; Night shift; Overtime pay; 4-6 rest days/month	Grade A sample
21	SF Ex-press	Yingkou	No night/rotating shift disclosed			Sorting; Loading/unloading	Grade A sample
22	SF Ex-press	Cangzhou	No night/rotating shift disclosed			Sorting; unloading truck s; sorting matrix; Small parcels; Forklift; SOP	Grade A sample
23	SF Ex-press	Haikou	No night/rotating shift disclosed			Sorting; Transfer hub; Handheld scanner; Conveyor belt; Automation equipment	Grade A sample
24	SF Ex-press	Nanping	No night/rotating shift disclosed; time window(s): 10-15			Sorting; Loading/unloading; Scanning; Handling; Bulky parcels; Warehouse	Grade A sample; Includes loading/unloading or bulky-parcel work where relevant
25	SF Ex-press	Chengdu	No night/rotating shift disclosed			Warehouse; Picking; Checking/rechecking; Packing; Outbound processing	Grade B sample
26	SF Ex-press	Nanjing	Night or rotating shift reported; time window(s): 19:30-06:30; 11 hours/day reported	11.0		Sorting; Scanning; Loading/unloading; Small parcels; Night shift	Grade B sample; Employer may be indirect, outsourced, platform-related or SF/JD-related rather than a direct entity
27	SF Ex-press	Shenzhen	No night/rotating shift disclosed			Logistics sorting	Grade B sample
28	SF Ex-press	Shanghai	No night/rotating shift disclosed			Pickup/delivery; Not pure sorting	Grade B sample; Not a pure sorting role; exclude from the main robot-vs-labour model
29	SF Ex-press	Guangzhou	No night/rotating shift disclosed			Not pure sorting; Needs supplementation	Grade B sample; Coverage gap or missing reliable city sample; Not a pure sorting role; exclude from the main robot-vs-labour model
30	SF Ex-press	Suzhou	No night/rotating shift disclosed			Coverage gap	Grade B sample; Coverage gap or missing reliable city sample
1	JD Logistics	Guangzhou	No night/rotating shift disclosed				Public listing sample; use with source URL for verification
2	JD Logistics	Wuhan	No night/rotating shift disclosed; time window(s): 9:00-21:00			Sorting / Picking / Stock handling / Checking/rechecking	Public listing sample; use with source URL for verification
3	JD Logistics	Wuhan	No night/rotating shift disclosed			Sorting / Scanning / Dispatching / Checking/rechecking	Public listing sample; use with source URL for verification
4	JD Logistics	Wuhan	No night/rotating shift disclosed			Packing / Picking / put-away/shelving / Stock handling / Receiving	Public listing sample; use with source URL for verification
5	JD Logistics	Wuhan	No night/rotating shift disclosed			Sorting / Packing / Picking / put-away/shelving / Receiving	Public listing sample; use with source URL for verification
6	JD Logistics	Wuhan	Night or rotating shift reported			Sorting / Packing / put-away/shelving / Stock handling / Loading/unloading / Forklift	Monthly equivalent was converted from hourly/daily wage assumptions; Includes loading/unloading or bulky-parcel work where relevant
7	JD Logistics	Guangzhou	No night/rotating shift disclosed			Sorting / Scanning	Public listing sample; use with source URL for verification
8	JD Logistics	Xingtai	Night or rotating shift reported			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
9	JD Logistics	Cangzhou	No night/rotating shift disclosed			Sorting	Public listing sample; use with source URL for verification

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No.	Brand	City	Shift summary	Hours/ day	Days/ month	Work content	Clean notes
10	JD Logistics	Hefei	Night or rotating shift reported			Sorting / Scanning	Public listing sample; use with source URL for verification
11	JD Logistics	Jingzhou	Night or rotating shift reported; time window(s): 17-05			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
12	JD Logistics	Wuhu	Night or rotating shift reported			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
13	JD Logistics	Tianjin	Night or rotating shift reported; time window(s): 7-20			Sorting / Dispatching	Public listing sample; use with source URL for verification
14	JD Logistics	Taiyuan	Night or rotating shift reported			Sorting / Loading/unloading	Monthly equivalent was converted from hourly/daily wage assumptions
15	JD Logistics	Tianjin	No night/rotating shift disclosed			Sorting / Loading/unloading	Monthly equivalent was converted from hourly/daily wage assumptions; Includes loading/unloading or bulky-parcel work where relevant
16	JD Logistics	Quanzhou	No night/rotating shift disclosed			Sorting / Loading/unloading / Dispatching	Includes loading/unloading or bulky-parcel work where relevant
17	JD Logistics	Zibo	Night or rotating shift reported			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
18	JD Logistics	Guangzhou	No night/rotating shift disclosed; time window(s): 8-22			Sorting / Packing / Picking	Temporary or short-term sample
19	JD Logistics	Wenzhou	Night or rotating shift reported			Sorting / Palletizing	Monthly equivalent was converted from hourly/daily wage assumptions
20	JD Logistics	Wuxi	No night/rotating shift disclosed			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
21	JD Logistics	Suzhou	Night or rotating shift reported			Sorting / Loading/unloading	Public listing sample; use with source URL for verification
22	JD Logistics	Xuzhou	Night or rotating shift reported			Sorting	Monthly equivalent was converted from hourly/daily wage assumptions
23	JD Logistics	Shanghai	No night/rotating shift disclosed			Sorting / Packing / Picking / Stock handling	Public listing sample; use with source URL for verification
24	JD Logistics	Dalian	Night or rotating shift reported			Sorting / Scanning / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
25	JD Logistics	Zhengzhou	Night or rotating shift reported			Sorting / Loading/unloading	Public listing sample; use with source URL for verification
26	JD Logistics	Shanghai	No night/rotating shift disclosed			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
27	JD Logistics	Hefei	Night or rotating shift reported			Sorting	Monthly equivalent was converted from hourly/daily wage assumptions
28	JD Logistics	Chengdu	No night/rotating shift disclosed			Sorting / Packing	Public listing sample; use with source URL for verification
29	JD Logistics	Guiyang	No night/rotating shift disclosed			Sorting / Loading/unloading	Includes loading/unloading or bulky-parcel work where relevant
30	JD Logistics	Wuxi	No night/rotating shift disclosed			Sorting / Packing / Picking / Receiving	Public listing sample; use with source URL for verification
31	JD Logistics	Ningbo	No night/rotating shift disclosed			Sorting / Packing / Exception handling	Public listing sample; use with source URL for verification
32	JD Logistics	Yanbian	No night/rotating shift disclosed			Sorting	Public listing sample; use with source URL for verification
33	JD Logistics	Ningbo	No night/rotating shift disclosed			Sorting / Scanning / Loading/unloading / Palletizing	Includes loading/unloading or bulky-parcel work where relevant
34	JD Logistics	Zhongshan	No night/rotating shift disclosed			Sorting	Public listing sample; use with source URL for verification
35	JD Logistics	Nantong	Night or rotating shift reported; time window(s): 18:00-6:00			Sorting	Public listing sample; use with source URL for verification
36	JD Logistics	Fuzhou	Night or rotating shift reported; time window(s): 7-7			Sorting / Stock handling	Public listing sample; use with source URL for verification
37	JD Logistics	Chengdu	Night or rotating shift reported			Sorting	Monthly equivalent was converted from hourly/daily wage assumptions
38	JD Logistics	Nanchang	No night/rotating shift disclosed			Sorting / Scanning / Stock handling / Loading/unloading	Temporary or short-term sample; Includes loading/unloading or bulky-parcel work where relevant
39	JD Logistics	Tianjin	Night or rotating shift reported			Sorting	Monthly equivalent was converted from hourly/daily wage assumptions
40	JD Logistics	Taizhou	Night or rotating shift reported			Sorting / Palletizing / Dispatching	Monthly equivalent was converted from hourly/daily wage assumptions
41	JD Logistics	Suzhou	No night/rotating shift disclosed			Sorting / Packing / Picking / Receiving	Temporary or short-term sample
42	JD Logistics	Guangzhou	No night/rotating shift disclosed			Sorting / Scanning / Loading/unloading / Palletizing	Includes loading/unloading or bulky-parcel work where relevant
43	JD Logistics	Urumqi	Night or rotating shift reported			Sorting	Public listing sample; use with source URL for verification
44	JD Logistics	Wuxi	Night or rotating shift reported			Sorting / Packing / Picking / Loading/unloading / Checking/rechecking / Stocktaking / Forklift	Includes loading/unloading or bulky-parcel work where relevant
45	JD Logistics	Luoyang	Night or rotating shift reported; time window(s): 19-07			Sorting / Scanning	Public listing sample; use with source URL for verification
46	JD Logistics	Guangzhou	No night/rotating shift disclosed			Sorting / Packing / Picking / Stock handling	Public listing sample; use with source URL for verification
47	JD Logistics	Chengdu	No night/rotating shift disclosed			Sorting / Scanning	Public listing sample; use with source URL for verification
48	JD Logistics	Nanning	Night or rotating shift reported			Sorting / Packing / Picking / Loading/unloading	Monthly equivalent was converted from hourly/daily wage assumptions; Includes loading/unloading or bulky-parcel work where relevant
49	JD Logistics	Guangzhou	No night/rotating shift disclosed			Sorting / Packing / Picking / put-away/shelving / Stock handling / Loading/unloading / Stocktaking	Includes loading/unloading or bulky-parcel work where relevant
50	JD Logistics	Zhuhai	No night/rotating shift disclosed			Sorting / Packing / Picking / Loading/unloading / Palletizing	Includes loading/unloading or bulky-parcel work where relevant
51	JD Logistics	Chengdu	No night/rotating shift disclosed			Packing / Picking	Public listing sample; use with source URL for verification
52	JD Logistics	Chengdu	No night/rotating shift disclosed			Packing / Picking / put-away/shelving / Stocktaking / Forklift	Public listing sample; use with source URL for verification
53	JD Logistics	Chengdu	No night/rotating shift disclosed			Packing / Picking / put-away/shelving / Stock handling / Checking/rechecking / Receiving	Public listing sample; use with source URL for verification
54	JD Logistics	Hangzhou	No night/rotating shift disclosed			Sorting / Packing / Picking / put-away/shelving / Stocktaking / Forklift	Public listing sample; use with source URL for verification
55	JD Logistics	Beijing	Night or rotating shift reported			Sorting / Loading/unloading	Temporary or short-term sample; Includes loading/unloading or bulky-parcel work where relevant
56	JD Logistics	Beijing	No night/rotating shift disclosed			Packing / Picking	Public listing sample; use with source URL for verification
57	JD Logistics	Beijing	No night/rotating shift disclosed			Packing / Picking	Public listing sample; use with source URL for verification

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No.	Brand	City	Shift summary	Hours/ day	Days/ month	Work content	Clean notes
58	JD Logistics	Shanghai	Night or rotating shift reported			Sorting / Scanning	Public listing sample; use with source URL for verification
59	JD Logistics	Shanghai	No night/rotating shift disclosed			Sorting / Scanning	Public listing sample; use with source URL for verification
60	JD Logistics	Shanghai	No night/rotating shift disclosed			Sorting / Palletizing	Public listing sample; use with source URL for verification
61	JD Logistics	Nanjing	No night/rotating shift disclosed			Packing / Picking / put-away/shelving / Stocktaking / Receiving	Public listing sample; use with source URL for verification
62	JD Logistics	Zhenjiang	No night/rotating shift disclosed			Packing / Picking / put-away/shelving / Stocktaking / Receiving	Public listing sample; use with source URL for verification
63	JD Logistics	Nanning	No night/rotating shift disclosed			Packing / Picking / put-away/shelving	Public listing sample; use with source URL for verification
64	JD Logistics	Hefei	No night/rotating shift disclosed			Packing / Picking / put-away/shelving / Checking/rechecking	Public listing sample; use with source URL for verification
65	JD Logistics	Nanchang	Night or rotating shift reported				Public listing sample; use with source URL for verification
66	JD Logistics	Chongqing	No night/rotating shift disclosed			Sorting / Loading/unloading / Dispatching	Includes loading/unloading or bulky-parcel work where relevant
67	JD Logistics	Zhuhai	No night/rotating shift disclosed			Sorting / Scanning	Public listing sample; use with source URL for verification
68	JD Logistics	Shenyang	No night/rotating shift disclosed			Sorting / Packing / Picking / put-away/shelving / Stock handling / Checking/rechecking / Stocktaking / Receiving	Public listing sample; use with source URL for verification

Role-level source and verification notes

Table A10: Role-level source and verification notes

No.	Brand	City	Job title	Source URL	Reliability / verification notes
1	SF Express	Beijing	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/da7e201085d546991XB-29y4EQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
2	SF Express	Hangzhou	SF Express High-pay Parcel Sorter	https://m.zhipin.com/zhaopin/134a3d24d67320bc1XR_3di_GA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
3	SF Express	Xiamen	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/7b7efb3f3577c4271n1y3968EQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
4	SF Express	Sanya	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/7b7efb3f3577c4271n1y3968EQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
5	SF Express	Changchun	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/7b7efb3f3577c4271n1y3968EQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
6	SF Express	Jingzhou	SF Express Part-time Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/7b7efb3f3577c4271n1y3968EQ--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
7	SF Express	Zhongshan	SF Express Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/dac6a4f0148b031333Jy2t--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
8	SF Express	Jinan	SF Express Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/dac6a4f0148b031333Jy2t--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
9	SF Express	Dongguan	SF Express Sorting/Forklift Worker	https://www.zhipin.com/zhaopin/dac6a4f0148b031333Jy2t--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
10	SF Express	Wuhan	SF Express Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/5b49e33aa34977251nRy2Nu6EQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
11	SF Express	Tianjin	SF Express Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/03e58ad93ab22c4clnd53ti7EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
12	SF Express	Guilin	SF Express Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5b49e33aa34977251nRy2Nu6EQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
13	SF Express	Nanchang	SF Express Temporary/short-term Sorting/Forklift Worker	https://m.zhipin.com/zhaopin/f339e7c1cbd49aee1nB_3ty7Fg--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
14	SF Express	Xingtai	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/fcb22f9235278b9f1Xd-3925Fg--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
15	SF Express	Hefei	SF Express Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/ccd6987800dc2b3d1nZ-3t--EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
16	SF Express	Weifang	SF Express Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/c7c84d9a33391a290HJ83N68/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
17	SF Express	Dalian	SF Express Part-time Parcel Sorter	https://m.zhipin.com/zhaopin/dcacf0919983bcl41nxz3Ni-EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
18	SF Express	Ningbo	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/8f23a17c4cab2b7833V90tm_/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
19	SF Express	Xi'an	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/dcacf0919983bcl41nxz3Ni-EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
20	SF Express	Taiyuan	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/dcacf0919983bcl41nxz3Ni-EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
21	SF Express	Yingkou	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/dcacf0919983bcl41nxz3Ni-EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.

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No.	Brand	City	Job title	Source URL	Reliability / verification notes
22	SF Express	Cangzhou	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/dcacf0919983bc141nxz3Ni-EA--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
23	SF Express	Haikou	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/8f23a17c4ca6b2b7833V90tm_/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
24	SF Express	Nanping	SF Express Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/08559a1e522ab2ac1nJy3Ny5GQ--/	Grade A; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
25	SF Express	Chengdu	SF Express Dispatch, Stock Handling, Picking and Shipping Worker	https://www.zhipin.com/zhaopin/5b49e33aa34977251nRy2Nu6EQ--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
26	SF Express	Nanjing	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/c32fcc5747157be11HF43t-6Eg--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
27	SF Express	Shenzhen	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/27bfc7a17060db371XZ63928Fw--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
28	SF Express	Shanghai	SF Express Parcel Sorter	https://m.zhipin.com/zhaopin/4291cab6d53e01cb1nB82NW0/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
29	SF Express	Guangzhou	SF Express Parcel Sorter	https://www.zhipin.com/zhaopin/c0b94db1dd6b32991XZ43Nu4FA--/	Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
30	SF Express	Suzhou	SF Express To be added		Grade B; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
1	JD Logistics	Guangzhou	JD Logistics Direct-hire Regular Warehouse Operator	https://m.lienpin.com/job/1963834335.shtml	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
2	JD Logistics	Wuhan	JD Logistics Sorting and Stock Handling Worker	https://www.zhipin.com/zhaopin/e6c6150e08bd22703J72N-1GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
3	JD Logistics	Wuhan	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/e6c6150e08bd22703J72N-1GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
4	JD Logistics	Wuhan	JD Logistics Direct-hire Picking and Packing Warehouse Worker	https://www.zhipin.com/zhaopin/e6c6150e08bd22703J72N-1GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
5	JD Logistics	Wuhan	JD Logistics Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/e6c6150e08bd22703J72N-1GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
6	JD Logistics	Wuhan	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
7	JD Logistics	Guangzhou	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
8	JD Logistics	Xingtai	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
9	JD Logistics	Cangzhou	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
10	JD Logistics	Hefei	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
11	JD Logistics	Jingzhou	JD Logistics Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
12	JD Logistics	Wuhu	JD Logistics Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
13	JD Logistics	Tianjin	JD Logistics Dispatch, Stock Handling, Picking and Shipping Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
14	JD Logistics	Taiyuan	JD Logistics Part-time Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
15	JD Logistics	Tianjin	JD Logistics Direct-hire Part-time High-pay Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
16	JD Logistics	Quanzhou	JD Logistics Regular Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
17	JD Logistics	Zibo	JD Logistics Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
18	JD Logistics	Guangzhou	JD Logistics Temporary/short-term Picking and Packing Warehouse Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
19	JD Logistics	Wenzhou	JD Logistics Direct-hire Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
20	JD Logistics	Wuxi	JD Logistics Direct-hire Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
21	JD Logistics	Suzhou	JD Logistics Regular Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.

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No.	Brand	City	Job title	Source URL	Reliability / verification notes
22	JD Logistics	Xuzhou	JD Logistics Direct-hire Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
23	JD Logistics	Shanghai	JD Logistics Sorting and Stock Handling Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
24	JD Logistics	Dalian	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
25	JD Logistics	Zhengzhou	JD Logistics Direct-hire Regular Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
26	JD Logistics	Shanghai	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
27	JD Logistics	Hefei	JD Logistics Direct-hire Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
28	JD Logistics	Chengdu	JD Logistics Regular Store/Front-warehouse Picking and Packing Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
29	JD Logistics	Guiyang	JD Logistics Direct-hire Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
30	JD Logistics	Wuxi	JD Logistics Regular Parcel Sorter	https://www.zhipin.com/zhaopin/5f49e8882ce5bdcc1nN93961GA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
31	JD Logistics	Ningbo	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
32	JD Logistics	Yanbian	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
33	JD Logistics	Ningbo	JD Logistics Sorting and Loading/Unloading Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
34	JD Logistics	Zhongshan	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
35	JD Logistics	Nantong	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
36	JD Logistics	Fuzhou	JD Logistics Sorting and Stock Handling Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
37	JD Logistics	Chengdu	JD Logistics Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
38	JD Logistics	Nanchang	JD Logistics Temporary/short-term Sorting and Stock Handling Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
39	JD Logistics	Tianjin	JD Logistics Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
40	JD Logistics	Taizhou	JD Logistics Dispatch, Stock Handling, Picking and Shipping Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
41	JD Logistics	Suzhou	JD Logistics Temporary/short-term Regular Picking and Packing Warehouse Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
42	JD Logistics	Guangzhou	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
43	JD Logistics	Urumqi	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
44	JD Logistics	Wuxi	JD Logistics Picking and Packing Warehouse Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
45	JD Logistics	Luoyang	JD Logistics Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
46	JD Logistics	Guangzhou	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
47	JD Logistics	Chengdu	JD Logistics Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
48	JD Logistics	Nanning	JD Logistics Picking and Packing Warehouse Worker	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
49	JD Logistics	Guangzhou	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/5a5fb5f7beeb13533nR52Nm4/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
50	JD Logistics	Zhuhai	JD Logistics Direct-hire Regular Warehouse Worker	https://www.zhipin.com/zhaopin/4d2c34069ea1ee841nF73di0FA--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
51	JD Logistics	Chengdu	JD Logistics Direct-hire Warehouse Worker	https://m.zhipin.com/zhaopin/702abe96ae56104a1HR60tu9Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.

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No.	Brand	City	Job title	Source URL	Reliability / verification notes
52	JD Logistics	Chengdu	JD Logistics Warehouse Operator	https://m.zhipin.com/zhaopin/702abe96ae56104a1HR60tu9Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
53	JD Logistics	Chengdu	JD Logistics Warehouse Worker	https://m.zhipin.com/zhaopin/702abe96ae56104a1HR60tu9Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
54	JD Logistics	Hangzhou	JD Logistics Direct-hire Regular Parcel Sorter	https://www.zhipin.com/zhaopin/365293ec87e185981nR72N8--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
55	JD Logistics	Beijing	JD Logistics Direct-hire Temporary/short-term Parcel Sorter	https://www.zhipin.com/zhaopin/0983387955962af51nJ529S1Ew--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
56	JD Logistics	Beijing	JD Logistics Store/Front-warehouse Picking and Packing Worker	https://www.zhipin.com/zhaopin/0983387955962af51nJ529S1Ew--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
57	JD Logistics	Beijing	JD Logistics Direct-hire Store/Front-warehouse Picking and Packing Worker	https://www.zhipin.com/zhaopin/0983387955962af51nJ529S1Ew--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
58	JD Logistics	Shanghai	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/f74a8fc3757a1fcd1nB93dy0Ew--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
59	JD Logistics	Shanghai	JD Logistics Part-time Parcel Sorter	https://www.zhipin.com/zhaopin/f74a8fc3757a1fcd1nB93dy0Ew--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
60	JD Logistics	Shanghai	JD Logistics Direct-hire Parcel Sorter	https://www.zhipin.com/zhaopin/f74a8fc3757a1fcd1nB93dy0Ew--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
61	JD Logistics	Nanjing	JD Logistics High-pay Warehouse Operator	https://www.zhipin.com/zhaopin/58e49bc8a7926aff1Xx_0t24Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
62	JD Logistics	Zhenjiang	JD Logistics Regular Warehouse Worker	https://www.zhipin.com/zhaopin/58e49bc8a7926aff1Xx_0t24Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
63	JD Logistics	Nanning	JD Logistics Warehouse Worker	https://www.zhipin.com/zhaopin/58e49bc8a7926aff1Xx_0t24Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
64	JD Logistics	Hefei	JD Logistics Direct-hire Warehouse Worker	https://www.zhipin.com/zhaopin/bbbe23448215cd3a1nNy2Nu1Eg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
65	JD Logistics	Nanchang	JD Logistics Direct-hire Warehouse Operator	https://www.zhipin.com/zhaopin/d682ac6f91249bd60H173NW9/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
66	JD Logistics	Chongqing	JD Logistics Dispatch, Stock Handling, Picking and Shipping Worker	https://www.zhipin.com/zhaopin/ca577581baf1ce4033B-3NW-/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
67	JD Logistics	Zhuhai	JD Logistics Parcel Sorter	https://www.zhipin.com/zhaopin/ca577581baf1ce4033B-3NW-/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.
68	JD Logistics	Shenyang	JD Logistics Warehouse Operator	https://www.zhipin.com/zhaopin/58e49bc8a7926aff1Xx_0t24Fg--/	Unrated; public recruitment/listing excerpt; recruitment pages are dynamic, so secondary verification is recommended before formal modelling.

Appendix B: Worker efficiency and sorting-quality parameters

This appendix documents the worker-efficiency and sorting-quality parameters used in the dynamic humanoid robot break-even simulation. Publicly observed values are reported where available. When no public worker-level value is available, the public-value field is reported as N/A and the model value is treated as a scenario assumption.

Table B1: Summary of base worker-efficiency inputs

pph = parcels per hour. Effective parcels per shift include effective working time and quality adjustment used in the simulation workbook.

Carrier	Base pph	Shift hours	Effective time	Peak loss	Effective parcels/shift	Basis
SF Express	500	10.5	80%	15%	3,213	Base model uses public 500 parcels/hour benchmark and recruitment-based shift evidence.
JD Logistics	400	10.0	80%	15%	2,448	Base model uses lower worker throughput due to lack of public JD worker-level sorting evidence.

Table B2: Indicator-level evidence classification

N/A means that no public worker-level value was available in the collected sources.

ID	Category	Variable	Carrier/context	Scope	Public evidence	Data status
SF_MANUAL_PPH	Productivity	Parcels processed per hour	SF Express / manual sorting	Worker-level comparison benchmark	Public comparison benchmark: manual sorting about 500 parcels/hour.	Public benchmark - media/non-official KPI
SF_AUTO_HUB_PPH	Productivity	Automated system throughput	SF South China operating hub	System/equipment level	Public system benchmark: automated sorting equipment processes at least 71,000 parcels/hour.	Public system-level data

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ID	Category	Variable	Carrier/context	Scope	Public evidence	Data status
SF_AUTOMATION_INVEST	Feasibility/automation investment	Transfer-hub automation investment	SF Express 2025H1	Company-level operating data	Public company material: transfer efficiency +12.5%; >120 transfer automation units; >380 AGV forklifts; automation capacity >5.57 million parcels/day.	Public company material
IND_HZ_DAILY	Productivity/quality	Large logistics-park daily capacity	Hangzhou Chengnan smart logistics park (industry benchmark)	Site/park level	Public industry benchmark: logistics park processes 2.2 million parcels/day with 99.99% sorting accuracy.	Public system-level benchmark - company not named
IND_HZ_ACC	Quality	Automated sorting accuracy	Hangzhou Chengnan smart logistics park (industry benchmark)	Site/park level	Public industry benchmark: automated sorting accuracy 99.99%.	Public system-level benchmark - company not named
SF_SHIFT	Shift	Shift length	SF Express / recruitment-based sorting sample	Recruitment/job-posting level	Recruitment evidence: 5h, 10h, 10.5h, 11h, 12h and 12.5h shifts appear in postings.	Public recruitment information - verification required
SF_EFFECTIVE_TIME	Productivity	Effective working-time ratio	SF Express / manual sorting	Model assumption	N/A (no public worker-level effective working-time ratio).	Model assumption
SF_WRONG_SORT	Quality	Manual wrong-sorting rate	SF Express / manual sorting	Model assumption	N/A (no public worker-level wrong-sorting rate).	Model assumption
SF_MISSCAN	Quality	Manual missed-scan rate	SF Express / manual sorting	Model assumption	N/A (no public worker-level missed-scan rate).	Model assumption
SF_DAMAGE	Quality	Damage rate	SF Express / manual sorting	Model assumption	N/A (no public worker-level damage rate).	Model assumption
SF_REWORK	Quality	Rework rate	SF Express / manual sorting	Model assumption	N/A (no public worker-level rework rate).	Model assumption
SF_PEAK	Peak	Peak-period human productivity loss	SF Express / manual sorting	Model assumption	N/A (no public worker-level peak productivity loss).	Model assumption
SF_NIGHT	Night shift	Night-shift productivity loss	SF Express / manual sorting	Model assumption	N/A (no public worker-level night-shift productivity loss).	Model assumption
JD_AUTO_KUNSHAN	Productivity	Unmanned sorting-centre throughput	JD Kunshan unmanned sorting centre	System/equipment level	N/A	Public system-level data
JD_AUTO_ASIA_DAILY	Productivity/quality	Kunshan Asia No.1 automated sorting centre	JD Kunshan Asia No.1 sorting centre	Site/system level	N/A	Public system-level data
JD_SORT_WALL_PPH	Productivity/quality	Automatic sorting-wall throughput	JD Logistics automatic sorting wall	Equipment level	N/A	Public equipment-level data
JD_PICKER_ASSIST	Productivity	Robot-assisted picking productivity	JD Logistics robot-assisted picking	Human-robot picking	N/A	Public human-robot collaboration data
JD_SMART_WAREHOUSE	Productivity/cost	Zhilang goods-to-person system productivity	JD Logistics smart warehousing	System/equipment level	N/A	Public system-level data
JD_WORKER_PPH	Productivity	Parcels processed per hour	JD Logistics / manual sorting	Model assumption	N/A	Model assumption
JD_SHIFT	Shift	Shift length	JD Logistics / recruitment-based sorting sample	Recruitment/job-posting level	Recruitment evidence: 4h, 6h, 10h, 12h shifts and night/two-shift patterns appear in postings.	Public recruitment information - verification required
JD_EFFECTIVE_TIME	Productivity	Effective working-time ratio	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level effective working-time ratio).	Model assumption
JD_WRONG_SORT	Quality	Manual wrong-sorting rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level wrong-sorting rate).	Model assumption
JD_MISSCAN	Quality	Manual missed-scan rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level missed-scan rate).	Model assumption
JD_DAMAGE	Quality	Damage rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level damage rate).	Model assumption
JD_REWORK	Quality	Rework rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level rework rate).	Model assumption
JD_PEAK	Peak	Peak-period human productivity loss	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level peak productivity loss).	Model assumption
JD_NIGHT	Night shift	Night-shift productivity loss	JD Logistics / manual sorting	Model assumption	N/A (no public worker-level night-shift productivity loss).	Model assumption

Table B3: Indicator-level numerical inputs and model use

Low, base and high values are scenario inputs. Public evidence and assumption status are reported in Table A2.

ID	Low	Base	High	Unit	Model use	Source
SF_MANUAL_PPH	300	500	600	parcels/hour/worker	Usable with benchmark caveat	Sohu / maker community repost of Qizhong Intelligence
SF_AUTO_HUB_PPH	71,000	71,000	71,000	parcels/hour/system	No: system-level only	Sohu / maker community repost of Qizhong Intelligence
SF_AUTOMATION_INVEST	12.5	12.5	12.5	% efficiency improvement	No: company-level context only	SF Holding 2025 interim investor presentation
IND_HZ_DAILY	2,200,000	2,200,000	2,200,000	parcels/day/site	No: system-level and company not named	CIMC Pteris / CIMC Deli logistics automation case
IND_HZ_ACC	100	100	100	%	No: system-level and company not named	CIMC Pteris / CIMC Deli logistics automation case
SF_SHIFT	8	10.5	12.5	hours/shift	Yes: adjust for effective working time	User-compiled BOSS Zhipin and Liepin recruitment table
SF_EFFECTIVE_TIME	70	80	85	%	Yes: sensitivity parameter	Model assumption based on recruitment shift patterns
SF_WRONG_SORT	0.1	0.5	1	%	Yes: model assumption	No public worker-level data; model assumption
SF_MISSCAN	0.1	0.5	1	%	Yes: model assumption	No public worker-level data; model assumption
SF_DAMAGE	0.05	0.2	0.5	%	Yes: model assumption	No public worker-level data; model assumption
SF_REWORK	0.5	1.5	3	%	Yes: model assumption	No public worker-level data; model assumption
SF_PEAK	10	15	25	% decline	Yes: model assumption	CIMC Deli industry description plus model assumption
SF_NIGHT	5	10	15	% decline	Yes: model assumption	User-compiled recruitment table plus model assumption
JD_AUTO_KUNSHAN	9,000	9,000	9,000	parcels/hour/system	No: system-level only	People's Daily Overseas Edition / People.cn
JD_AUTO_ASIA_DAILY	4,500,000	4,500,000	4,500,000	parcels/day/site	No: system-level only	CIMC Group news release
JD_SORT_WALL_PPH	1,800	1,800	1,800	parcels/hour/equipment	No: equipment-level only	Shanghai Securities News / Sohu repost
JD_PICKER_ASSIST	250	250	250	orders/hour/worker	Partly usable: different operating scope	JD Logistics news centre / People's Daily report
JD_SMART_WAREHOUSE	120	120	120	boxes/hour/robot	No: system/warehouse-picking scope	Science and Technology Daily
JD_WORKER_PPH	250	400	500	parcels/hour/worker	Yes: model assumption	Composite assumption from JD public picking evidence and SF manual benchmark
JD_SHIFT	4	10	12	hours/shift	Yes: adjust for effective working time	User-compiled BOSS Zhipin and Liepin recruitment table
JD_EFFECTIVE_TIME	70	80	85	%	Yes: sensitivity parameter	Model assumption based on recruitment shift patterns
JD_WRONG_SORT	0.1	0.5	1	%	Yes: model assumption	No public worker-level data; model assumption
JD_MISSCAN	0.1	0.5	1	%	Yes: model assumption	No public worker-level data; model assumption
JD_DAMAGE	0.05	0.2	0.5	%	Yes: model assumption	No public worker-level data; model assumption
JD_REWORK	0.5	1.5	3	%	Yes: model assumption	No public worker-level data; model assumption

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ID	Low	Base	High	Unit	Model use	Source
JD_PEAK	10	15	25	% decline	Yes: model assumption	JD public peak/automation materials plus model assumption
JD_NIGHT	5	10	15	% decline	Yes: model assumption	User-compiled recruitment table plus model assumption

Table B4: Indicator-level modelling basis and recommended use

This table separates observed benchmarks from estimated worker-level parameters.

ID	Modelling basis	Recommended use
SF_MANUAL_PPH	Used as a public benchmark, not as an audited internal KPI.	Use 300/500/600 pph as low/base/high scenarios.
SF_AUTO_HUB_PPH	System-level benchmark; not directly comparable with one worker or one humanoid robot.	Use as automation upper-bound context; do not compare with a single humanoid robot.
SF_AUTOMATION_INVOLV	Used for automation context and seasonal-capacity assumptions.	Use for background and peak-capacity assumptions.
IND_HZ_DAILY	Used as industry-level benchmark only.	Use for automation-maturity and peak-scenario background.
IND_HZ_ACC	Used as industry-level benchmark only.	Use as an optimistic accuracy benchmark for robot scenarios.
SF_SHIFT	Used for shift and night-work assumptions; not used as direct productivity evidence.	Use 10-12 hours for formal workers; model part-time roles separately.
SF_EFFECTIVE_TIME	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 70/80/85% as low/base/high scenarios.
SF_WRONG_SORT	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
SF_MISSCAN	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
SF_DAMAGE	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.05/0.2/0.5% for sensitivity analysis.
SF_REWORK	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.5/1.5/3.0% for sensitivity analysis.
SF_PEAK	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - peak loss) in peak scenarios.
SF_NIGHT	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - night-shift loss) in night scenarios.
JD_AUTO_KUNSHAN	System-level benchmark; not directly comparable with one worker or one humanoid robot.	Use for automation-replacement context; not a single-robot value.
JD_AUTO_ASIA_DAILY	System-level benchmark; not directly comparable with one worker or one humanoid robot.	Use as JD automation upper-bound and peak-capacity context.
JD_SORT_WALL_PPH	Equipment-level benchmark; not a worker-level observation.	Use as comparison for small/standard parcel automation.
JD_PICKER_ASSIST	Warehouse-picking benchmark; used only as indirect evidence.	Use as warehouse picking evidence, not direct parcel-sorting evidence.
JD_SMART_WAREHOUSE	System-level benchmark; not directly comparable with one worker or one humanoid robot.	Use as human-robot and robotic-system efficiency context.
JD_WORKER_PPH	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 250/400/500 pph as low/base/high scenarios.
JD_SHIFT	Used for shift and night-work assumptions; not used as direct productivity evidence.	Use 10-12 hours for formal workers; model 4-6 hour part-time shifts separately.
JD_EFFECTIVE_TIME	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 70/80/85% as low/base/high scenarios.
JD_WRONG_SORT	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
JD_MISSCAN	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
JD_DAMAGE	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.05/0.2/0.5% for sensitivity analysis.
JD_REWORK	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.5/1.5/3.0% for sensitivity analysis.
JD_PEAK	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - peak loss) in peak scenarios.
JD_NIGHT	No public worker-level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - night-shift loss) in night scenarios.

Table B5: Parcel-type manual productivity assumptions

Parcel-type efficiencies are model assumptions. They should be replaced by PDA, scanner or WMS/TMS logs when available.

Parcel type	Carrier	Low	Base	High	Unit	Status
Documents	SF Express	450	600	700	parcels/hour/worker	Model assumption - field validation required
Small standard parcels	SF Express	400	550	650	parcels/hour/worker	Model assumption - field validation required
Standard cartons	SF Express	300	450	550	parcels/hour/worker	Model assumption - field validation required
Soft parcels	SF Express	220	350	450	parcels/hour/worker	Model assumption - field validation required
Irregular parcels	SF Express	120	220	320	parcels/hour/worker	Model assumption - field validation required
Heavy/large parcels	SF Express	80	150	250	parcels/hour/worker	Model assumption - field validation required
Documents	JD Logistics	405	540	630	parcels/hour/worker	Model assumption - field validation required
Small standard parcels	JD Logistics	360	495	585	parcels/hour/worker	Model assumption - field validation required
Standard cartons	JD Logistics	270	405	495	parcels/hour/worker	Model assumption - field validation required
Soft parcels	JD Logistics	198	315	405	parcels/hour/worker	Model assumption - field validation required

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Parcel type	Carrier	Low	Base	High	Unit	Status
Irregular parcels	JD Logistics	108	198	288	parcels/hour/worker	Model assumption - field validation required
Heavy/large parcels	JD Logistics	72	135	225	parcels/hour/worker	Model assumption - field validation required

Table B6: Parcel-type robotic suitability assumptions

Task coverage is the estimated share of the parcel type that a humanoid robot can process reliably under the model assumptions.

Parcel type	Carrier	Robot suitability	Task coverage	Error-risk multiplier	Basis	Source/notes
Documents	SF Express	High	0.85	0.7	Documents are light and regular; they are expected to have high manual and automation productivity.	https://ir.sf-express.com/media/lokf2io2/2025-inter-im-investor-presentation-c.pdf
Small standard parcels	SF Express	Very high	0.8	0.8	Small parcels are relatively standardised and frequently appear in automation cases.	https://ir.sf-express.com/media/lokf2io2/2025-inter-im-investor-presentation-c.pdf ; https://www.sohu.com/a/685639813_120988576
Standard cartons	SF Express	High	0.65	1	Cartons are regular but vary in size and weight; they are suitable as the main model scenario.	Model assumption
Soft parcels	SF Express	Medium	0.4	1.3	Soft parcels deform and create unstable label and grasping positions.	Model assumption
Irregular parcels	SF Express	Medium-low	0.25	1.8	Irregular parcels often require extra review or specialised handling.	https://ir.sf-express.com/media/lokf2io2/2025-inter-im-investor-presentation-c.pdf
Heavy/large parcels	SF Express	Medium-low	0.2	2	Heavy and large parcels reduce manual productivity and increase damage or safety risk.	User uploaded file: Pasted text(265).txt / BOSS-Liepin extraction in this conversation
Documents	JD Logistics	High	0.85	0.7	Documents are light and regular; they are expected to have high manual and automation productivity.	https://ir.sf-express.com/media/lokf2io2/2025-inter-im-investor-presentation-c.pdf
Small standard parcels	JD Logistics	Very high	0.8	0.8	Small parcels are relatively standardised and frequently appear in automation cases.	https://ir.sf-express.com/media/lokf2io2/2025-inter-im-investor-presentation-c.pdf ; https://www.sohu.com/a/685639813_120988576
Standard cartons	JD Logistics	High	0.65	1	Cartons are regular but vary in size and weight; they are suitable as the main model scenario.	Model assumption
Soft parcels	JD Logistics	Medium	0.4	1.3	Soft parcels deform and create unstable label and grasping positions.	Model assumption
Irregular parcels	JD Logistics	Medium-low	0.25	1.8	Irregular parcels often require extra review or specialised handling.	https://ir.sf-express.com/media/lokf2io2/2025-inter-im-investor-presentation-c.pdf
Heavy/large parcels	JD Logistics	Medium-low	0.2	2	Heavy and large parcels reduce manual productivity and increase damage or safety risk.	User uploaded file: Pasted text(265).txt / BOSS-Liepin extraction in this conversation

Table B7: Scenario-level worker productivity inputs

These scenario values feed the dynamic price-efficiency frontier and hybrid staffing simulation.

Carrier	Scenario	pph	Shift hours	Effective time	Effective parcels/shift	Basis
SF Express	Conservative	300	8	70.0%	1,071	Fatigue, peak pressure and exception items.
SF Express	Base	500	10.5	80.0%	3,213	Uses the 500 pph public benchmark and recruitment shift evidence.
SF Express	Optimistic	600	12	85.0%	5.23e+03	Skilled workers, standard parcels and non-peak conditions.
JD Logistics	Conservative	250	8	70.0%	892	No public worker-level JD sorting value; low case informed by public robot-assisted picking evidence.
JD Logistics	Base	400	10	80.0%	2,448	Recruitment-based shift assumption and lower than SF 500 pph benchmark.
JD Logistics	Optimistic	500	12	85.0%	4.36e+03	Standard small parcels, skilled workers and non-peak conditions.

Table B8: Scenario-level quality and productivity-loss factors

These values are assumptions where no public worker-level value is available; N/A is used in the public evidence table for missing observed worker-level data.

Carrier	Scenario	Night loss	Peak loss	Wrong-sort	Missed-scan	Damage	Rework
SF Express	Conservative	15.0%	25.0%	1.0%	1.0%	0.5%	3.0%
SF Express	Base	10.0%	15.0%	0.5%	0.5%	0.2%	1.5%
SF Express	Optimistic	5.0%	10.0%	0.1%	0.1%	0.1%	0.5%
JD Logistics	Conservative	15.0%	25.0%	1.0%	1.0%	0.5%	3.0%
JD Logistics	Base	10.0%	15.0%	0.5%	0.5%	0.2%	1.5%
JD Logistics	Optimistic	5.0%	10.0%	0.1%	0.1%	0.1%	0.5%

Table B9: Source overview

Reliability refers to public-source usefulness for this model, not to audited operational truth.

Source ID	Source	Supported data	Reliability
S1	Sohu / maker community repost: SF automated sorting system	SF South China hub automatic sorting throughput of 71,000 parcels/hour and a 500 parcels/hour manual comparison benchmark.	Medium

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Source ID	Source	Supported data	Reliability
S2	SF Holding 2025 interim investor presentation	Transfer efficiency +12.5%; more than 120 new transfer automation units; more than 380 new AGV forklifts; automated capacity above 5.57 million parcels/day.	High
S3	CIMC Deli Hangzhou Chengnan smart logistics park case	Daily capacity of 2.2 million parcels; sorting accuracy 99.99%; peak promotion volume up to three times normal level.	Medium
S4	People's Daily Overseas Edition / People.cn: JD Kunshan unmanned sorting centre	9,000 parcels/hour; 180 workers saved under comparable site and volume conditions; 10-minute charge for 4 hours of work.	High
S5	CIMC Group: JD Kunshan Asia No.1 phase II	More than 80 automated sorting lines; 10,000 intelligent sorting robots; 99.99% accuracy; more than 4.5 million parcels/day during 618; fivefold efficiency improvement.	Medium-high
S6	Shanghai Securities News / Sohu: JD Logistics automatic sorting wall	Automatic sorting wall: 1,800 parcels/hour; scanning/operation accuracy 99.99%; efficiency improvement 50%.	Medium
S7	JD Logistics news centre / People's Daily report	Robot-assisted picking: about 250 orders/hour per picker; about three times the traditional mode.	Medium-high
S8	Science and Technology Daily: JD Zhi-lang goods-to-person system	One flying-ladder robot handles about 120 boxes/hour; outbound picking efficiency +300%; clothing order-level logistics cost -50%.	Medium-high
S9	User-compiled BOSS Zhipin and Liepin recruitment table	Recruitment evidence on shift length, night work, part-time work, 12-hour shifts and wages for SF and JD sorting roles.	Medium
E1	Model assumption	Effective working rate, wrong-sorting, missed scans, damage, rework, night-shift loss and peak-period loss.	Low-medium

Table B10: Source URLs and use limitations

URLs are retained for auditability. Recruitment extracts and model assumptions should be updated when internal logs become available.

Source ID	URL or file	Use limitation
S1	https://www.sohu.com/a/206009074_194357	Media repost rather than official SF KPI; used as a public benchmark.
S2	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf	Company material; useful for automation context, not worker-level productivity.
S3	https://cn.cimc-pterisglobal.com/slider/cimc-pteris-showcases-logistics-automation-solutions-in-hangzhou-chengnan/	Company not named; used as an industry benchmark.
S4	https://paper.people.com.cn/rmrhwb/html/2018-05/30/content_1857991.htm	Authoritative media report; system-level data only.
S5	https://www.cimc.com/index.php?a=show&c=index&catid=17&id=3142&m=content	Supplier press release; system-level data.
S6	https://www.sohu.com/a/685639813_120988576	Equipment/case-level data; not equivalent to manual parcel sorting.
S7	https://www.jingdonglogistics.com/cn/news/200	Warehouse picking order data; not parcel sorting throughput.
S8	https://www.stdaily.com/web/gdxw/2025-08/01/content_379195.html	Warehouse picking system data, not manual parcel sorting.
S9	User uploaded file: Pasted text(265).txt / BOSS-Liepin extraction in this conversation	Public recruitment extracts; postings change over time and require verification.
E1	No direct public worker-level data found	Must be reported as assumptions and replaced with company logs or field observations when available.

Appendix C. Humanoid robot price, BOM and total ownership cost data

This appendix reports the robot-price, bill-of-materials (BOM), cost-category and total-ownership-cost inputs used in the humanoid robot parcel-sorting economic model. All monetary values are retained in the currency used in the input workbook unless explicitly stated. Missing or unconfirmed values are reported as N/A. Tables are split into portrait-format blocks to preserve readability and avoid horizontal overflow.

Table C1: Workbook overview for humanoid robot cost appendix

Summary values taken from the dashboard sheet of the robot cost workbook.

Item	Value
Workbook purpose	Track 2026 public prices and cost/TCO buckets for humanoid robots relevant to parcel sorting ROI.
Data status	Mix of official public prices, media-reported prices, and enterprise estimates. Quote vendors before final investment decision.
Models covered	18
Lowest public price RMB	9,998
Median public price RMB	180,000
Highest public price RMB	1,800,000

Table C2: Dashboard summary of robot models and five-year TCO

This table summarizes the robot models retained for cost comparison. TCO values include deployment, operating and depreciation assumptions from the workbook.

ID	Company	Model	Price mid RMB	5-year TCO RMB	Comment
1	Noetix Robotics	Bumi / Xiaobumi	9,998	468,316.94	Low-cost/dev class
2	Unitree	R1 Air	35,280	506,998.4	Low-cost/dev class
3	Unitree	R1	42,480	518,014.4	Low-cost/dev class
4	Unitree	R1 domestic entry	33,400	504,122	Low-cost/dev class
5	Booster Robotics	Booster K1	29,900	498,767	Low-cost/dev class
6	Astriobot	T1	89,900	590,567	Low-cost/dev class
7	Unitree	G1	97,200	601,736	Low-cost/dev class
8	Unitree	H2	215,280	782,398.4	Potential pilot class
9	Unitree	H1	648,000	1,444,460	High-cost enterprise class
10	Unitree	H2 Plus	720,000	1,554,620	High-cost enterprise class
11	AgiBot / Zhiyuan	Lingxi X2 Youth	98,000	602,960	Low-cost/dev class
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	183,000	733,010	Potential pilot class

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ID	Company	Model	Price mid RMB	5-year TCO RMB	Comment
13	AgiBot / Zhiyuan	Genie G1	450,000	1,141,520	High-cost enterprise class
14	UBTECH	Walker S2	676,800	1,488,524	High-cost enterprise class
15	Fourier Intelligence	GR-1	1,584,000	2,876,540	Very high-cost enterprise/RaaS
16	Agility Robotics	Digit	1,800,000	3,207,020	Very high-cost enterprise/RaaS
17	Tesla	Optimus	180,000	728,420	Potential pilot class

Table C3: Robot price database: model identity and price status

Public-price status separates official prices, media-reported prices, enterprise estimates and benchmark-only values.

ID	Company	Model	Segment	Region	Price mid RMB	Price status	Quality
1	Noetix Robotics	Bumi / Xiaobumi	Consumer / education	China	9,998	Public sale price	High
2	Unitree	R1 Air	Entry humanoid	China	35,280	Official shop / product page	High
3	Unitree	R1	Entry humanoid	China	42,480	Official shop / product page	High
4	Unitree	R1 domestic entry	Entry humanoid	China	33,400	Media-reported domestic range	Medium
5	Booster Robotics	Booster K1	Education / research	China	29,900	Public launch price	High
6	AstriBot	T1	Industrial / general	China	89,900	Media-reported starting price	Medium
7	Unitree	G1	Research / development	China	97,200	Official shop price	High
8	Unitree	H2	Full-size humanoid	China	215,280	Official shop price	High
9	Unitree	H1	Full-size humanoid	China	648,000	Official shop indicative price	High
10	Unitree	H2 Plus	Full-size humanoid	China	720,000	Official shop price	High
11	AgiBot / Zhiyuan	Lingxi X2 Youth	Education / interaction	China	98,000	Public sale price	High
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	Interaction / service	China	183,000	Public sale / promo range	High
13	AgiBot / Zhiyuan	Genie G1	High-end humanoid	China	450,000	Public sale price	High
14	UBTECH	Walker S2	Industrial humanoid	China	676,800	Enterprise estimate / quote	Low
15	Fourier Intelligence	GR-1	Industrial / research	China	1,584,000	Partner quote	Medium
16	Agility Robotics	Digit	Warehouse logistics	United States	1,800,000	Industry estimate / RaaS	Medium
17	Tesla	Optimus	Benchmark / not for sale	United States	180,000	Target price; not purchasable	Low
18	Tesla	Optimus Gen 2 BOM benchmark	BOM benchmark	United States / China supply chain	637,200	BOM estimate range	Medium

Table C4: Robot price database: original price ranges and capability notes

Original prices are retained to show whether the public value was reported in RMB, USD or another currency.

ID	Model	Original low	Original high	Curr.	RMB low	RMB high	Use case / capability
1	Bumi / Xiaobumi	9,998	9,998	RMB	9,998	9,998	94cm, ~12kg, 21+ DOF; education/companion/developer use
2	R1 Air	4,900	4,900	USD	35,280	35,280	Compact humanoid, 20 DOF, tax/shipping excluded
3	R1	5,900	5,900	USD	42,480	42,480	Compact humanoid, 26 DOF, tax/shipping excluded
4	R1 domestic entry	26,900	39,900	RMB	26,900	39,900	Domestic 2026 mass-market price reference
5	Booster K1	29,900	29,900	RMB	29,900	29,900	1m, ~19kg, developer/education platform
6	T1	89,900	89,900	RMB	89,900	89,900	Industrial humanoid entering sub-RMB 100k range
7	G1	13,500	13,500	USD	97,200	97,200	Compact 35kg humanoid, 23-43 joint motors, optional dexterous hand
8	H2	29,900	29,900	USD	215,280	215,280	Full-size humanoid listed on Unitree shop
9	H1	90,000	90,000	USD	648,000	648,000	Full-size H1, contact sales for real price
10	H2 Plus	100,000	100,000	USD	720,000	720,000	Higher-end H2 Plus listed on Unitree shop
11	Lingxi X2 Youth	98,000	98,000	RMB	98,000	98,000	1.3m-class interaction/development humanoid
12	Yuanzheng A2 Youth	168,000	198,000	RMB	168,000	198,000	1.7m-class interaction/service humanoid
13	Genie G1	450,000	450,000	RMB	450,000	450,000	High-end embodied intelligence product
14	Walker S2	68,000	120,000	USD	489,600	864,000	Industrial factory/logistics model with hot-swap battery
15	GR-1	220,000	220,000	USD	1,584,000	1,584,000	Full-size research/rehab/industrial pilot humanoid

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ID	Model	Original low	Original high	Curr.	RMB low	RMB high	Use case / capability
16	Digit	250,000	250,000	USD	1,800,000	1,800,000	Warehouse bin/tote handling; Amazon pilots; often RaaS
17	Optimus	20,000	30,000	USD	144,000	216,000	Future scaled-production price target only
18	Optimus Gen 2 BOM benchmark	46,000	131,000	USD	331,200	943,200	China supply-chain cost vs ex-China supply-chain cost estimate

Table C5: Robot price database: source URLs and source notes

URLs are retained as plain text links for auditability. Some enterprise prices are estimates or quotes and should be verified before procurement decisions.

ID	Company	Model	Source URL	Notes
1	Noetix Robotics	Bumi / Xiaobumi	https://noetixrobotics.com/news/media/213	Not a logistics-grade sorter; price useful as lower-bound consumer benchmark.
2	Unitree	R1 Air	https://www.unitree.com/mobile/R1/	Official page lists R1 Air at \$4,900.
3	Unitree	R1	https://www.unitree.com/mobile/R1/	Official page lists R1 at \$5,900; domestic media also reports lower local entry prices.
4	Unitree	R1 domestic entry	https://www.chinadailyasia.com/article/634870	China Daily Asia reports starting price RMB 26,900; earlier Reuters/media reported ~RMB 39,900.
5	Booster Robotics	Booster K1	https://www.chinadailyasia.com/article/634870	China Daily Asia reports K1 series starting price RMB 29,900.
6	Astriobot	T1	https://www.chinadailyasia.com/article/634870	Reported starting price; full configuration may be higher.
7	Unitree	G1	https://shop.unitree.com/collections/humanoid-robot	Official shop lists G1 at \$13,500; Chinese media reports G1 base near RMB 85k.
8	Unitree	H2	https://shop.unitree.com/collections/humanoid-robot	Official shop lists H2 at \$29,900.
9	Unitree	H1	https://shop.unitree.com/collections/humanoid-robot	Official shop shows \$90,000 and says contact for real price.
10	Unitree	H2 Plus	https://shop.unitree.com/collections/humanoid-robot	Official shop lists H2 Plus at \$100,000.
11	AgiBot / Zhiyuan	Lingxi X2 Youth	https://news.cnyes.com/news/id/6115374	Reported as AgiBot's cheapest humanoid model at RMB 98,000.
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	https://news.cnyes.com/news/id/6115374	Cnyes reports RMB 198,000 list and RMB 30,000 limited discount; Sina reports RMB 168,000.
13	AgiBot / Zhiyuan	Genie G1	https://finance.sina.com.cn/tech/roll/2025-08-18/doc-infmmmy1961486.shtml	Sina/Jiemian reports Genie G1 at RMB 450,000.
14	UBTECH	Walker S2	https://blog.robozaps.com/b/ubtech-walker-s-review	UBTECH enterprise quotes vary; public exact price not disclosed.
15	Fourier Intelligence	GR-1	https://www.roboticscenter.ai/store/product/fourier-gr1/	SVRC partner quote lists \$220,000; official Fourier page requires quote.
16	Agility Robotics	Digit	https://blog.robozaps.com/b/agility-robotics-digit-review	Estimated outright price; Agility often sells service/subscription deployment.
17	Tesla	Optimus	https://standardbots.com/blog/tesla-robot	No confirmed retail price; use only as forecast/benchmark, not current purchase quote.
18	Tesla	Optimus Gen 2 BOM benchmark	https://wallstreetcn.com/articles/3774890	WallstreetCN cites Tesla BOM \$46k using China supply chain vs \$131k using US/non-China supply chain.

Table C6: BOM benchmark components and cost values

BOM values combine public/component benchmarks from BofA, Roland Berger and WallstreetCN-cited component estimates.

Benchmark	Source	Component	Share	Cost low	Cost high	Curr.	RMB low	RMB high
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Linear actuator integration	0.27	9,450	9,450	USD	68,040	68,040
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Rotary actuator integration	0.24	8,400	8,400	USD	60,480	60,480
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Dexterous hand / end effector	0.19	6,650	6,650	USD	47,880	47,880
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	AI chip / AI computing	0.1	3,500	3,500	USD	25,200	25,200
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Vision system	0.04	1,400	1,400	USD	10,080	10,080
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Battery	0.02	700	700	USD	5,040	5,040
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Heat management system	0.02	700	700	USD	5,040	5,040
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Other body / structural / connectors	0.12	4,200	4,200	USD	30,240	30,240
Roland Berger 2026 entry-level component model	Roland Berger	Motion – actuator	N/A	1,265	1,265	USD	9,108	9,108

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Benchmark	Source	Component	Share	Cost low	Cost high	Curr.	RMB low	RMB high
Roland Berger 2026 entry-level component model	Roland Berger	Hand/end-effector system	N/A	333	333	USD	2,397.6	2,397.6
Roland Berger 2026 entry-level component model	Roland Berger	Motion – other	N/A	342	342	USD	2,462.4	2,462.4
Roland Berger 2026 entry-level component model	Roland Berger	Energy & charging	N/A	380	380	USD	2,736	2,736
Roland Berger 2026 entry-level component model	Roland Berger	Skeleton & structural components	N/A	820	820	USD	5,904	5,904
Roland Berger 2026 entry-level component model	Roland Berger	Compute & connectivity	N/A	345	345	USD	2,484	2,484
Roland Berger 2026 entry-level component model	Roland Berger	Perception system	N/A	166	166	USD	1,195.2	1,195.2
Roland Berger 2026 entry-level component model	Roland Berger	Others	N/A	230	230	USD	1,656	1,656
Roland Berger 2026 advanced component model	Roland Berger	Motion – actuator	N/A	4,000	4,000	USD	28,800	28,800
Roland Berger 2026 advanced component model	Roland Berger	Hand/end-effector system	N/A	1,400	1,400	USD	10,080	10,080
Roland Berger 2026 advanced component model	Roland Berger	Motion – other	N/A	1,072	1,072	USD	7,718.4	7,718.4
Roland Berger 2026 advanced component model	Roland Berger	Energy & charging	N/A	845	845	USD	6,084	6,084
Roland Berger 2026 advanced component model	Roland Berger	Skeleton & structural components	N/A	1,950	1,950	USD	14,040	14,040
Roland Berger 2026 advanced component model	Roland Berger	Compute & connectivity	N/A	800	800	USD	5,760	5,760
Roland Berger 2026 advanced component model	Roland Berger	Perception system	N/A	465	465	USD	3,348	3,348
Roland Berger 2026 advanced component model	Roland Berger	Others	N/A	440	440	USD	3,168	3,168
Unitree G1 teardown / media estimate	WallstreetCN / Zhineng Auto	Total material cost / BOM	N/A	41,600	41,600	RMB	41,600	41,600
Unitree G1 teardown / media estimate	WallstreetCN / Zhineng Auto	Core joints	N/A	27,500	27,500	RMB	27,500	27,500
Tesla Optimus supply-chain estimate	WallstreetCN / Morgan Stanley estimate cited	BOM with Chinese supply chain	N/A	46,000	46,000	USD	331,200	331,200
Tesla Optimus supply-chain estimate	WallstreetCN / Morgan Stanley estimate cited	BOM with US/non-China supply chain	N/A	131,000	131,000	USD	943,200	943,200

Table C7: BOM benchmark notes and source links

Notes describe the component-level assumptions behind each cost estimate.

Benchmark	Component	Notes	Source URL
BofA typical humanoid BOM (China-made components, end-2025)	Linear actuator integration	Assumes 14 linear actuators based on planetary roller screws.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Rotary actuator integration	Assumes 16 rotary actuators with harmonic reducers.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Dexterous hand / end effector	Assumes hand with 6 DOF.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	AI chip / AI computing	Onboard compute / AI chip.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Vision system	One depth camera and one LiDAR assumption.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Battery	Battery pack.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Heat management system	Thermal management.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Other body / structural connectors	Residual percentage to make total 100%.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf

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Benchmark	Component	Notes	Source URL
Roland Berger 2026 entry-level component model	Motion – actuator	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Hand/end-effector system	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Motion – other	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Energy & charging	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Skeleton & structural components	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Compute & connectivity	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Perception system	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Others	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Motion – actuator	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Hand/end-effector system	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Motion – other	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Energy & charging	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Skeleton & structural components	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Compute & connectivity	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Perception system	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Others	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Unitree G1 teardown / media estimate	Total material cost / BOM	Media article states G1 material cost RMB 41,600.	https://wallstreetcn.com/articles/3774890
Unitree G1 teardown / media estimate	Core joints	Media article states G1 core joints RMB 27,500.	https://wallstreetcn.com/articles/3774890
Tesla Optimus supply-chain estimate	BOM with Chinese supply chain	Media article cites \$46k with Chinese supply-chain components.	https://wallstreetcn.com/articles/3774890
Tesla Optimus supply-chain estimate	BOM with US/non-China supply chain	Media article cites \$131k if excluding Chinese components.	https://wallstreetcn.com/articles/3774890

Table C8: Robot TCO model: upfront deployment components

Upfront deployment cost adds the bare price, estimated parcel-handling adaptation, barcode/scanning, safety and integration assumptions.

ID	Company	Model	Price mid RMB	Hardware BOM RMB	End effector	Barcode	Safety	Integration	Upfront cost
1	Noetix Robotics	Bumi / Xi-aobumi	9,998	5,998.8	999.8	5,000	299.94	1,999.6	18,297.34
2	Unitree	R1 Air	35,280	21,168	3,528	5,000	1,058.4	7,056	51,922.4
3	Unitree	R1	42,480	25,488	4,248	5,000	1,274.4	8,496	61,498.4
4	Unitree	R1 domestic entry	33,400	20,040	3,340	5,000	1,002	6,680	49,422
5	Booster Robotics	Booster K1	29,900	17,940	2,990	5,000	897	5,980	44,767
6	Astriobot	T1	89,900	53,940	8,990	5,000	2,697	17,980	124,567
7	Unitree	G1	97,200	58,320	9,720	5,000	2,916	19,440	134,276
8	Unitree	H2	215,280	129,168	21,528	5,000	6,458.4	43,056	291,322.4
9	Unitree	H1	648,000	388,800	64,800	5,000	19,440	129,600	866,840
10	Unitree	H2 Plus	720,000	432,000	72,000	5,000	21,600	144,000	962,600
11	AgiBot / Zhiyuan	Lingxi X2 Youth	98,000	58,800	9,800	5,000	2,940	19,600	135,340
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	183,000	109,800	18,300	5,000	5,490	36,600	248,390
13	AgiBot / Zhiyuan	Genie G1	450,000	270,000	45,000	5,000	13,500	90,000	603,500
14	UBTECH	Walker S2	676,800	406,080	67,680	5,000	20,304	135,360	905,144
15	Fourier Intelligence	GR-1	1,584,000	950,400	158,400	5,000	47,520	316,800	2,111,720
16	Agility Robotics	Digit	1,800,000	1,080,000	180,000	5,000	54,000	360,000	2,399,000
17	Tesla	Optimus	180,000	108,000	18,000	5,000	5,400	36,000	244,400

Table C9: Robot TCO model: annual operating components*Annual operating costs are modelled per robot and are used to convert robot price into annual economic cost.*

ID	Model	Maintenance	Software	Ops personnel	Electricity	Downtime loss
1	Bumi / Xiaobumi	799.84	28,800	22,500	2,304	36,000
2	R1 Air	2,822.4	28,800	22,500	2,304	36,000
3	R1	3,398.4	28,800	22,500	2,304	36,000
4	R1 domestic entry	2,672	28,800	22,500	2,304	36,000
5	Booster K1	2,392	28,800	22,500	2,304	36,000
6	T1	7,192	28,800	22,500	2,304	36,000
7	G1	7,776	28,800	22,500	2,304	36,000
8	H2	17,222.4	28,800	22,500	2,304	36,000
9	H1	51,840	28,800	22,500	2,304	36,000
10	H2 Plus	57,600	28,800	22,500	2,304	36,000
11	Lingxi X2 Youth	7,840	28,800	22,500	2,304	36,000
12	Yuanzheng A2 Youth	14,640	28,800	22,500	2,304	36,000
13	Genie G1	36,000	28,800	22,500	2,304	36,000
14	Walker S2	54,144	28,800	22,500	2,304	36,000
15	GR-1	126,720	28,800	22,500	2,304	36,000
16	Digit	144,000	28,800	22,500	2,304	36,000
17	Optimus	14,400	28,800	22,500	2,304	36,000

Table C10: Robot TCO model: depreciation, residual value and five-year TCO*Five-year TCO is the cumulative ownership-cost benchmark used for comparing low-cost, pilot-class and enterprise-class humanoids.*

ID	Model	Residual value	Annual depreciation	Annual operating cost	5-year TCO	TCO / purchase	Status / comment
1	Bumi / Xiaobumi	1,999.6	3,259.55	90,403.84	468,316.94	46.84	Low-cost/dev class
2	R1 Air	7,056	8,973.28	92,426.4	506,998.4	14.37	Low-cost/dev class
3	R1	8,496	10,600.48	93,002.4	518,014.4	12.19	Low-cost/dev class
4	R1 domestic entry	6,680	8,548.4	92,276	504,122	15.09	Low-cost/dev class
5	Booster K1	5,980	7,757.4	91,996	498,767	16.68	Low-cost/dev class
6	T1	17,980	21,317.4	96,796	590,567	6.57	Low-cost/dev class
7	G1	19,440	22,967.2	97,380	601,736	6.19	Low-cost/dev class
8	H2	43,056	49,653.28	106,826.4	782,398.4	3.63	Potential pilot class
9	H1	129,600	147,448	141,444	1,444,460	2.23	High-cost enterprise class
10	H2 Plus	144,000	163,720	147,204	1,554,620	2.16	High-cost enterprise class
11	Lingxi X2 Youth	19,600	23,148	97,444	602,960	6.15	Low-cost/dev class
12	Yuanzheng A2 Youth	36,600	42,358	104,244	733,010	4.01	Potential pilot class
13	Genie G1	90,000	102,700	125,604	1,141,520	2.54	High-cost enterprise class
14	Walker S2	135,360	153,956.8	143,748	1,488,524	2.20	High-cost enterprise class
15	GR-1	316,800	358,984	216,324	2,876,540	1.82	Very high-cost enterprise/RaaS
16	Digit	360,000	407,800	233,604	3,207,020	1.78	Very high-cost enterprise/RaaS
17	Optimus	36,000	41,680	104,004	728,420	4.05	Potential pilot class

Table C11: Robot cost-category map: definitions and default estimation*The cost-category map links the model's requested cost buckets to concrete data items and estimation rules.*

Cost bucket	What it includes	Default estimation	Use in ROI	Caveat
Bare machine / robot body	Public robot purchase price or manufacturer quote; excludes tax/shipping/custom integration	Use mid public price from Robot_Price_DB	Main capex; compare against annual labour saving	Public prices often refer to developer models, not industrial configurations.
End effector	Dexterous hands, gripper, suction cup, parcel handling tooling, tactile sensors	End_effector_upgrade_rate x hardware price	Essential for parcels; default hands may not be enough for soft packs and irregular parcels	Often optional and not included in base robot price.
Vision system	Depth camera, LiDAR, RGB cameras, industrial camera, lighting, OCR vision	Included in base if manufacturer includes; otherwise allocate from BOM benchmarks	Required for parcel localisation and bin-picking	Can be much higher if adding industrial cameras and lighting.
Barcode/scanning system	Fixed scanner, handheld/robot-mounted 1D/2D scanner, OCR/DWS integration	Scanner_system_base_RMB	Critical for express sorting; humanoid vision alone does not replace barcode workflows	Usually not part of humanoid price.
Mobility system	Leg actuators, joint modules, reducers, motors, screws, encoders, bearings, servo drives	Part of base robot; compare to BOM actuator shares	Determines whether robot can move between stations or only stand at risk fixed workcell	Largest cost bucket, also main maintenance risk.
Safety system	E-stop, collision avoidance, protective covers, fences, warning lights, risk assessment, insurance enablement	Safety_system_rate_base x hardware price	Required if operating near workers; affects deployment permission	Often hidden in integration budget.
Software cost	Fleet software, robot SDK, navigation updates, cloud, monitoring, VLA/LLM subscriptions	Software_license_base_RMB per robot/year	Can determine whether robot is Opex-heavy even if hardware is cheap	Vendors may bundle first-year software.
Integration cost	WMS/WCS/TMS integration, workflow redesign, simulation, training, acceptance testing	Integration_rate_base x hardware price	Often decides whether pilot can scale	Highly site-specific; real quote needed.

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Cost bucket	What it includes	Default estimation	Use in ROI	Caveat
Maintenance cost	Preventive maintenance, spare parts, battery replacement, joint/gear wear, service contract	Maintenance_rate_base x hardware price/year	Heavy-duty night sorting may push maintenance upward	Warehouse robotics benchmark; humanoid data still limited.
Operations personnel	Robot technician, trainer, remote operator, fleet supervisor	Ops_engineer_annual_cost_RMB / Robots_per_ops_engineer	If one person manages few robots, ROI worsens sharply	Supervision need is often underestimated.
Electricity	Charging energy and battery swap station power	Power_kW x hours/day x days/year x tariff	Small compared with labour, but useful for complete TCO	Power varies by task intensity and payload.
Downtime loss	Lost throughput, backup labour, missed cut-off, service delay and recovery costs	Downtime_hours_per_year x Downtime_cost_per_hour_RMB	Important for express logistics because cut-off times matter	Hard to estimate without site data.
Depreciation	Annual accounting cost of robot, integration and accessories net of residual value	Straight-line depreciation over Useful_life_years	Turns capex into annual cost to compare with annual labour cost	Rapid price declines can shorten economic life.
Residual value	Expected resale/scrap value at end of useful life	Residual_rate x hardware price	Higher residual improves payback; low residual raises risk	Very uncertain in 2026 due to rapid price declines.

Table C12: Robot cost-category map: data collection and low/base/high logic

These fields specify what should be collected from vendors or site pilots to replace modelling proxies.

Cost bucket	Recommended data to collect	Low/base/high logic
Bare machine / robot body	Model, configuration, hand option, battery option, warranty, tax, shipping	If no quote, use low/high public price range
End effector	Hand model, payload, grip success rate, parcel types, replacement frequency	5% / 10% / 20% of hardware price
Vision system	Sensor bill, camera resolution, lighting, perception software	BofA vision 4%; Roland perception entry \$166/advanced \$465
Barcode/scanning system	Scanner unit price, mounting, SDK/WMS integration, scan success rate	RMB 500 / RMB 5,000 / RMB 20,000
Mobility system	DOF, actuator count, motor/reducer supplier, payload, walking speed, uptime	BofA linear+rotary actuator = 51%; Wall-streetCN says joint modules ~half
Safety system	CE/GB compliance, safety certification, risk assessment cost, near-miss events	1% / 3% / 8% of hardware price
Software cost	Annual license, API usage, cloud compute, model updates, cybersecurity	RMB 7,200 / RMB 28,800 / RMB 72,000 per year
Integration cost	Engineering days, WMS API, safety testing, line down-time during installation	10% / 20% / 50% of hardware price
Maintenance cost	Service contract, warranty, replacement schedule, critical spares, MTBF	5% / 8% / 12% of hardware price/year
Operations personnel	Technician salary, robots per technician, teleoperation need	Depends on supervision ratio
Electricity	kWh per hour, operating hours, electricity tariff	Editable power and hours
Downtime loss	Downtime hours, parcel throughput/hour, penalty/backup labour cost/hour	Editable low/base/high
Depreciation	Useful life, residual value, capex basis	3 / 5 / 7 years
Residual value	Second-hand market, supplier buyback, spare-part value	5% / 20% / 35%

Table C13: Robot cost assumptions and low/base/high values

Core cost assumptions are explicitly separated from the robot price database so they can be modified in sensitivity analysis.

Parameter	Base	Low	High	Unit
USD_to_RMB	7.2	7	7.4	RMB per USD
EUR_to_RMB	7.8	7.5	8.1	RMB per EUR
Maintenance_rate_base	0.08	0.05	0.12	% of hardware price/year
Software_license_base_RMB	28,800	7,200	72,000	RMB/year
Integration_rate_base	0.2	0.1	0.5	% of hardware price
Safety_system_rate_base	0.03	0.01	0.08	% of hardware price
Scanner_system_base_RMB	5,000	500	20,000	RMB upfront
End_effector_upgrade_rate	0.1	0.05	0.2	% of hardware price
Ops_engineer_annual_cost_RMB	180,000	120,000	300,000	RMB/year
Robots_per_ops_engineer	8	4	15	robots/person
Electricity_tariff_RMB_per_kWh	0.8	0.6	1.2	RMB/kWh
Power_kW_base	0.6	0.3	1.2	kW
Operating_hours_per_day	16	8	22	hours/day
Operating_days_per_year	300	250	350	days/year
Downtime_hours_per_year	120	40	300	hours/year
Downtime_cost_per_hour_RMB	300	100	1,000	RMB/hour
Useful_life_years	5	3	7	years
Residual_rate	0.2	0.05	0.35	% of initial hardware price
BOM_proxy_rate_of_sale_price	0.6	0.4	0.8	%

Table C14: Robot cost assumptions: notes and source links

Source notes include URLs when the assumption was calibrated from public warehouse-robotics or humanoid-cost information.

Parameter	Notes / source URL
USD_to_RMB	Editable conversion assumption; keep explicit rather than hidden FX.
EUR_to_RMB	Editable conversion assumption.
Maintenance_rate_base	Warehouse robotics guide suggests 5-12% of hardware cost annually: https://robotomated.com/learn/cost/warehouse-robot-cost-guide
Software_license_base_RMB	Proxy converted from \$1,000-\$10,000 per robot/year: https://robotomated.com/learn/cost/warehouse-robot-cost-guide
Integration_rate_base	Site-specific WMS/WCS integration, tooling, safety testing; editable proxy.

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Parameter	Notes / source URL
Safety_system_rate_base	Safety barriers, E-stop, warning lights, insurance enablement; not usually included in robot price.
Scanner_system_base_RMB	Barcode/OCR scanner add-on for logistics; industrial modules range from low-cost modules to rugged scanners.
End_effector_upgrade_rate	For parcel sorting, gripper/suction/dexterous-hand upgrade or replacement; editable proxy.
Ops_engineer_annual_cost_RMB	One robotics technician/engineer fully loaded annual cost assumption.
Robots_per_ops_engineer	Supervision ratio; increase after deployment maturity.
Electricity_tariff_RMB_per_kWh	Editable industrial/commercial tariff assumption.
Power_kW_base	Proxy based on 0.5-2 kWh per charge patterns and humanoid battery/runtime data; adjust by model/task.
Operating_hours_per_day	For logistics night/day shift scenario.
Operating_days_per_year	Editable site utilisation assumption.
Downtime_hours_per_year	Maintenance/unplanned downtime proxy.
Downtime_cost_per_hour_RMB	Throughput loss + backup labour + delay penalty proxy.
Useful_life_years	Straight-line depreciation assumption.
Residual_rate	Residual value highly uncertain because humanoid prices are falling quickly.
BOM_proxy_rate_of_sale_price	If no public BOM: estimate hardware BOM as share of public selling price; disclose as proxy only.

Appendix D: Baseline human labour cost inputs

Table D1: Baseline human labour cost inputs

Input	SF Express	JD Logistics
Average monthly wage (USD)	948.33	1,000.49
Annual employer labour cost (USD/FTE/year)	14,794.00	15,628.79
Human sorting throughput (parcels/hour)	500	400
Quality adjusted annual throughput (parcels/FTE/year)	975,834	743,493
Human sorting cost (USD/parcel)	0.01516	0.02102

Appendix E: Detailed result

Dashboard

Table E1: Dashboard - main threshold results

Summary block extracted from the Dashboard sheet.

Main threshold result for Unitree G1 at \$13,500	SF Express	JD Logistics
Normal season: required pph for unit-cost break-even	336.1	242.4
Normal season: required pph for 3-year payback	394.4	284.5
At q=300 pph: max G1 price for 3-year payback	4,112	15,644
At q=400 pph: max G1 price for 3-year payback	14,056	29,432
Conclusion	For SF, G1 must reach ~395 pph or cost <= ~\$4.1k at 300 pph for a 3-year payback. At 400 pph, \$13.5k becomes acceptable.	For JD, G1 is viable at 300 pph because the 3-year payback price ceiling is ~\$15.6k.

Table E2: Dashboard - seasonal G1 required throughput

Seasonal thresholds for the Unitree G1 benchmark.

Brand	Season	Demand/day	Human UC (\$/parcel)	q_BE unit	q_BE 3yr	Interpretation
SF Express	Cold/off-peak	50,000	0.0131	363.3	426.2	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
SF Express	Normal	100,000	0.0152	336.1	394.4	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
SF Express	Hot/peak	200,000	0.0214	292.4	343.1	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.

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Brand	Season	Demand/day	Human UC (\$/parcel)	q_BE unit	q_BE 3yr	Interpretation
JD Logistics	Cold/off-peak	50,000	0.0181	262	307.4	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
JD Logistics	Normal	100,000	0.0210	242.4	284.5	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
JD Logistics	Hot/peak	200,000	0.0297	210.9	247.4	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.

Parameters

Table E3: Parameters

All monetary values are in USD. The exchange-rate assumption is retained from the earlier model and can be edited.

Parameter	Value	Unit	Role in model	Source / rationale	Column 6
FX_CNY_per_USD	6.78	CNY/USD	Converts salary and cost data	Editable assumption from earlier USD workbook	
Workdays per year	312	days	Annual capacity conversion	26 days/month x 12	
Robot useful life	5	years	Depreciation	Base deployment assumption	
Residual rate	0.20	share of bare price	Depreciation	Base residual assumption	
Target payback	3	years	Investment break-even	Commercial investment threshold	
End-effector upgrade rate	0.10	share of bare price	Upfront robot deployment	Gripper/suction/dexterous-hand adaptation	
Safety system rate	0.0300	share of bare price	Upfront robot deployment	E-stop/protection/safety deployment	
Integration rate	0.20	share of bare price	Upfront robot deployment	WMS/WCS integration and site engineering	
Barcode/OCR system	737.4	USD upfront	Upfront robot deployment	Scanner/OCR proxy	
Maintenance rate	0.0800	share of bare price/year	Annual robot OPEX	Warehouse robotics maintenance proxy	
Annual software	4,247	USD/year	Annual robot OPEX	Robot software/cloud proxy	
Annual operations personnel allocation	3,318	USD/year	Annual robot OPEX	Robotics technician allocation	
Annual electricity	339.8	USD/year	Annual robot OPEX	Charging/electricity proxy	
Annual downtime loss	5,309	USD/year	Annual robot OPEX	Downtime-loss proxy	
Robot cost coefficient a	0.31	USD/year per USD price	$C_R(P)=aP+b$	Derived from depreciation and maintenance	
Robot fixed annual cost b	13,362	USD/year	$C_R(P)=aP+b$	Derived from scanner depreciation and fixed OPEX	
Human input	SF Express	JD Logistics	Unit	Source / rationale	
Annual employer cost	14,794	15,629	USD/FTE/year	Derived from uploaded salary component table	
Base human unit cost	0.0152	0.0210	USD/parcel	Cost divided by quality-adjusted annual parcels	
Base annual good parcels	975,834	743,493	parcels/FTE/year	Derived from efficiency workbook	
Base monthly mid-point	948.3	1,000	USD/month	Derived from recruitment salary data	
Season	Demand/day	Human capacity multiplier	Robot capacity multiplier	Labour cost multiplier	Robotable parcel share
Cold/off-peak	50,000	1.10	1	0.95	0.75
Normal	100,000	1	1	1	0.70
Hot/peak	200,000	0.85	0.95	1.20	0.60

Interactive Calculator

Table E4: Interactive_Calculator

Edit the four input cells in column B to test a dynamic robot price-efficiency case.

Input	Value	Unit	Description
Robot bare price P	13,500	USD	Edit this value
Robot throughput q	300	parcels/hour	Edit this value

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Input	Value	Unit	Description
Carrier	SF Express		Use SF Express or JD Logistics
Season	Normal		Use Cold/off-peak, Normal, or Hot/peak
Output	Formula result	Unit	Decision interpretation
Human unit cost in selected season	0.0152	USD/parcel	
Robot annual output multiplier	3,433	annual parcels per pph	
Robot annual cost	17,493	USD/year	$C_R(P)=0.306P+13,362.44$
Robot annual output	1,029,820	parcels/year	q multiplied by seasonal robot multiplier
Robot unit cost	0.0170	USD/parcel	Lower than human UC means unit-cost efficient
Unit-cost efficient?	No		
Required q for unit-cost break-even	336.1	parcels/hour	Current q must be at least this value
Required q for 3-year payback	394.4	parcels/hour	More conservative investment criterion
Maximum price for unit-cost break-even at current q	7,353	USD	
Maximum price for 3-year pay-back at current q	4,112	USD	
Lookup table used by calculator			
Season	SF Express Human UC	JD Logistics Human UC	Robot annual multiplier
Cold/off-peak	0.0131	0.0182	3,678
Normal	0.0152	0.0210	3,433
Hot/peak	0.0214	0.0297	2,795

G1 Thresholds

Table E5: G1_Thresholds

This table answers: at Unitree G1 price = \$13,500, what throughput is needed?

Brand	Season	Demand/day	Seasonal human UC	q for unit-cost break-even	q for 3-year pay-back	Main managerial reading	Criterion
SF Express	Cold/off-peak	50,000	0.0131	363.3	426.2	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
SF Express	Normal	100,000	0.0152	336.1	394.4	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
SF Express	Hot/peak	200,000	0.0214	292.4	343.1	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
JD Logistics	Cold/off-peak	50,000	0.0182	262.0	307.4	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
JD Logistics	Normal	100,000	0.0210	242.4	284.5	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
JD Logistics	Hot/peak	200,000	0.0297	210.9	247.4	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold

Price Efficiency Frontier

Table E6: Price_Efficiency_Frontier - column block 1

For each throughput q, the table reports the maximum bare robot price that can break even.

Brand	Season	Throughput q (pph)	Pmax unit-cost (\$)	Pmax 3-year pay-back (\$)	G1 price (\$)	G1 feasible by unit cost?	G1 feasible by 3-year payback?
SF Express	Cold/off-peak	200	-12,194	-7,318	13,500	No	No
SF Express	Cold/off-peak	250	-4,326	-2,717	13,500	No	No
SF Express	Cold/off-peak	300	3,543	1,884	13,500	No	No
SF Express	Cold/off-peak	350	11,411	6,485	13,500	No	No
SF Express	Cold/off-peak	400	19,280	11,085	13,500	Yes	No
SF Express	Cold/off-peak	450	27,148	15,686	13,500	Yes	Yes
SF Express	Cold/off-peak	500	35,017	20,287	13,500	Yes	Yes
SF Express	Cold/off-peak	600	50,754	29,489	13,500	Yes	Yes
SF Express	Cold/off-peak	700	66,491	38,690	13,500	Yes	Yes
SF Express	Cold/off-peak	800	82,228	47,892	13,500	Yes	Yes
SF Express	Normal	200	-9,654	-5,833	13,500	No	No
SF Express	Normal	250	-1,151	-860.6	13,500	No	No

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Brand	Season	Throughput q (pph)	Pmax unit-cost (\$)	Pmax 3-year pay-back (\$)	G1 price (\$)	G1 feasible by unit cost?	G1 feasible by 3-year payback?
SF Express	Normal	300	7,353	4,112	13,500	No	No
SF Express	Normal	350	15,856	9,084	13,500	Yes	No
SF Express	Normal	400	24,360	14,056	13,500	Yes	Yes
SF Express	Normal	450	32,863	19,028	13,500	Yes	Yes
SF Express	Normal	500	41,367	24,000	13,500	Yes	Yes
SF Express	Normal	600	58,374	33,944	13,500	Yes	Yes
SF Express	Normal	700	75,381	43,888	13,500	Yes	Yes
SF Express	Normal	800	92,388	53,833	13,500	Yes	Yes
SF Express	Hot/peak	200	-4,566	-2,858	13,500	No	No
SF Express	Hot/peak	250	5,209	2,858	13,500	No	No
SF Express	Hot/peak	300	14,985	8,574	13,500	Yes	No
SF Express	Hot/peak	350	24,760	14,290	13,500	Yes	Yes
SF Express	Hot/peak	400	34,536	20,006	13,500	Yes	Yes
SF Express	Hot/peak	450	44,311	25,721	13,500	Yes	Yes
SF Express	Hot/peak	500	54,087	31,437	13,500	Yes	Yes
SF Express	Hot/peak	600	73,637	42,869	13,500	Yes	Yes
SF Express	Hot/peak	700	93,188	54,301	13,500	Yes	Yes
SF Express	Hot/peak	800	112,739	65,732	13,500	Yes	Yes
JD Logistics	Cold/off-peak	200	-27.4	-203.9	13,500	No	No
JD Logistics	Cold/off-peak	250	10,883	6,175	13,500	No	No
JD Logistics	Cold/off-peak	300	21,793	12,555	13,500	Yes	No
JD Logistics	Cold/off-peak	350	32,703	18,934	13,500	Yes	Yes
JD Logistics	Cold/off-peak	400	43,613	25,313	13,500	Yes	Yes
JD Logistics	Cold/off-peak	450	54,523	31,693	13,500	Yes	Yes
JD Logistics	Cold/off-peak	500	65,434	38,072	13,500	Yes	Yes
JD Logistics	Cold/off-peak	600	87,254	50,831	13,500	Yes	Yes
JD Logistics	Cold/off-peak	700	109,074	63,589	13,500	Yes	Yes
JD Logistics	Cold/off-peak	800	130,895	76,348	13,500	Yes	Yes
JD Logistics	Normal	200	3,494	1,855	13,500	No	No
JD Logistics	Normal	250	15,285	8,750	13,500	Yes	No
JD Logistics	Normal	300	27,076	15,644	13,500	Yes	Yes
JD Logistics	Normal	350	38,866	22,538	13,500	Yes	Yes
JD Logistics	Normal	400	50,657	29,432	13,500	Yes	Yes
JD Logistics	Normal	450	62,448	36,326	13,500	Yes	Yes
JD Logistics	Normal	500	74,238	43,220	13,500	Yes	Yes
JD Logistics	Normal	600	97,820	57,009	13,500	Yes	Yes
JD Logistics	Normal	700	121,401	70,797	13,500	Yes	Yes
JD Logistics	Normal	800	144,982	84,585	13,500	Yes	Yes
JD Logistics	Hot/peak	200	10,549	5,980	13,500	No	No
JD Logistics	Hot/peak	250	24,103	13,906	13,500	Yes	Yes
JD Logistics	Hot/peak	300	37,658	21,831	13,500	Yes	Yes
JD Logistics	Hot/peak	350	51,212	29,756	13,500	Yes	Yes
JD Logistics	Hot/peak	400	64,766	37,682	13,500	Yes	Yes
JD Logistics	Hot/peak	450	78,320	45,607	13,500	Yes	Yes
JD Logistics	Hot/peak	500	91,875	53,533	13,500	Yes	Yes
JD Logistics	Hot/peak	600	118,983	69,383	13,500	Yes	Yes
JD Logistics	Hot/peak	700	146,092	85,234	13,500	Yes	Yes
JD Logistics	Hot/peak	800	173,200	101,085	13,500	Yes	Yes

Table E7: Price_Efficiency_Frontier - column block 2

Continuation of Price_Efficiency_Frontier; first columns are repeated as row identifiers.

Brand	Season	Throughput q (pph)	Comment
SF Express	Cold/off-peak	200	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	250	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	300	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	350	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	400	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	450	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	500	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	600	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	700	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	800	3-year payback is the stricter frontier.
SF Express	Normal	200	3-year payback is the stricter frontier.
SF Express	Normal	250	3-year payback is the stricter frontier.
SF Express	Normal	300	3-year payback is the stricter frontier.
SF Express	Normal	350	3-year payback is the stricter frontier.
SF Express	Normal	400	3-year payback is the stricter frontier.
SF Express	Normal	450	3-year payback is the stricter frontier.
SF Express	Normal	500	3-year payback is the stricter frontier.
SF Express	Normal	600	3-year payback is the stricter frontier.
SF Express	Normal	700	3-year payback is the stricter frontier.

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Brand	Season	Throughput q (pph)	Comment
SF Express	Normal	800	3-year payback is the stricter frontier.
SF Express	Hot/peak	200	3-year payback is the stricter frontier.
SF Express	Hot/peak	250	3-year payback is the stricter frontier.
SF Express	Hot/peak	300	3-year payback is the stricter frontier.
SF Express	Hot/peak	350	3-year payback is the stricter frontier.
SF Express	Hot/peak	400	3-year payback is the stricter frontier.
SF Express	Hot/peak	450	3-year payback is the stricter frontier.
SF Express	Hot/peak	500	3-year payback is the stricter frontier.
SF Express	Hot/peak	600	3-year payback is the stricter frontier.
SF Express	Hot/peak	700	3-year payback is the stricter frontier.
SF Express	Hot/peak	800	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	200	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	250	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	300	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	350	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	400	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	450	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	500	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	600	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	700	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	800	3-year payback is the stricter frontier.
JD Logistics	Normal	200	3-year payback is the stricter frontier.
JD Logistics	Normal	250	3-year payback is the stricter frontier.
JD Logistics	Normal	300	3-year payback is the stricter frontier.
JD Logistics	Normal	350	3-year payback is the stricter frontier.
JD Logistics	Normal	400	3-year payback is the stricter frontier.
JD Logistics	Normal	450	3-year payback is the stricter frontier.
JD Logistics	Normal	500	3-year payback is the stricter frontier.
JD Logistics	Normal	600	3-year payback is the stricter frontier.
JD Logistics	Normal	700	3-year payback is the stricter frontier.
JD Logistics	Normal	800	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	200	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	250	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	300	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	350	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	400	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	450	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	500	3-year payback is the stricter frontier.

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Brand	Season	Throughput q (pph)	Comment
JD Logistics	Hot/peak	600	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	700	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	800	3-year payback is the stricter frontier.

Simulation Grid

Table E8: Simulation_Grid - column block 1

Each row solves the cost-minimizing integer mix of robots and human workers under demand and robotable-share constraints.

Brand	Season	Demand/day	Price P (\$)	Throughput q (pph)	Robot UC (\$/parcel)	Human UC seasonal	Unit-cost feasible?
SF Express	Cold/off-peak	50,000	5,000	250	0.0162	0.0131	No
SF Express	Cold/off-peak	50,000	5,000	300	0.0135	0.0131	No
SF Express	Cold/off-peak	50,000	5,000	350	0.0116	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	400	0.0101	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	500	0.0081	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	600	0.0067	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	800	0.0051	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	250	0.0179	0.0131	No
SF Express	Cold/off-peak	50,000	10,000	300	0.0149	0.0131	No
SF Express	Cold/off-peak	50,000	10,000	350	0.0128	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	400	0.0112	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	500	0.0089	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	600	0.0074	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	800	0.0056	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	250	0.0190	0.0131	No
SF Express	Cold/off-peak	50,000	13,500	300	0.0159	0.0131	No
SF Express	Cold/off-peak	50,000	13,500	350	0.0136	0.0131	No
SF Express	Cold/off-peak	50,000	13,500	400	0.0119	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	500	0.0095	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	600	0.0079	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	800	0.0059	0.0131	Yes
SF Express	Cold/off-peak	50,000	20,000	250	0.0212	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	300	0.0177	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	350	0.0151	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	400	0.0132	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	500	0.0106	0.0131	Yes
SF Express	Cold/off-peak	50,000	20,000	600	0.0088	0.0131	Yes
SF Express	Cold/off-peak	50,000	20,000	800	0.0066	0.0131	Yes
SF Express	Cold/off-peak	50,000	30,000	250	0.0245	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	300	0.0204	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	350	0.0175	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	400	0.0153	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	500	0.0123	0.0131	Yes
SF Express	Cold/off-peak	50,000	30,000	600	0.0102	0.0131	Yes
SF Express	Cold/off-peak	50,000	30,000	800	0.0077	0.0131	Yes
SF Express	Cold/off-peak	50,000	50,000	250	0.0312	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	300	0.0260	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	350	0.0223	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	400	0.0195	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	500	0.0156	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	600	0.0130	0.0131	Yes
SF Express	Cold/off-peak	50,000	50,000	800	0.0097	0.0131	Yes
SF Express	Normal	100,000	5,000	250	0.0174	0.0152	No
SF Express	Normal	100,000	5,000	300	0.0145	0.0152	Yes
SF Express	Normal	100,000	5,000	350	0.0124	0.0152	Yes
SF Express	Normal	100,000	5,000	400	0.0108	0.0152	Yes
SF Express	Normal	100,000	5,000	500	0.0087	0.0152	Yes
SF Express	Normal	100,000	5,000	600	0.0072	0.0152	Yes
SF Express	Normal	100,000	5,000	800	0.0054	0.0152	Yes
SF Express	Normal	100,000	10,000	250	0.0191	0.0152	No
SF Express	Normal	100,000	10,000	300	0.0159	0.0152	No
SF Express	Normal	100,000	10,000	350	0.0137	0.0152	Yes
SF Express	Normal	100,000	10,000	400	0.0120	0.0152	Yes
SF Express	Normal	100,000	10,000	500	0.0096	0.0152	Yes
SF Express	Normal	100,000	10,000	600	0.0080	0.0152	Yes
SF Express	Normal	100,000	10,000	800	0.0060	0.0152	Yes
SF Express	Normal	100,000	13,500	250	0.0204	0.0152	No
SF Express	Normal	100,000	13,500	300	0.0170	0.0152	No
SF Express	Normal	100,000	13,500	350	0.0146	0.0152	Yes
SF Express	Normal	100,000	13,500	400	0.0127	0.0152	Yes
SF Express	Normal	100,000	13,500	500	0.0102	0.0152	Yes
SF Express	Normal	100,000	13,500	600	0.0085	0.0152	Yes
SF Express	Normal	100,000	13,500	800	0.0064	0.0152	Yes
SF Express	Normal	100,000	20,000	250	0.0227	0.0152	No
SF Express	Normal	100,000	20,000	300	0.0189	0.0152	No
SF Express	Normal	100,000	20,000	350	0.0162	0.0152	No
SF Express	Normal	100,000	20,000	400	0.0142	0.0152	Yes
SF Express	Normal	100,000	20,000	500	0.0114	0.0152	Yes
SF Express	Normal	100,000	20,000	600	0.0095	0.0152	Yes
SF Express	Normal	100,000	20,000	800	0.0071	0.0152	Yes
SF Express	Normal	100,000	30,000	250	0.0263	0.0152	No
SF Express	Normal	100,000	30,000	300	0.0219	0.0152	No
SF Express	Normal	100,000	30,000	350	0.0188	0.0152	No
SF Express	Normal	100,000	30,000	400	0.0164	0.0152	No
SF Express	Normal	100,000	30,000	500	0.0131	0.0152	Yes
SF Express	Normal	100,000	30,000	600	0.0109	0.0152	Yes
SF Express	Normal	100,000	30,000	800	0.0082	0.0152	Yes
SF Express	Normal	100,000	50,000	250	0.0334	0.0152	No
SF Express	Normal	100,000	50,000	300	0.0278	0.0152	No
SF Express	Normal	100,000	50,000	350	0.0239	0.0152	No
SF Express	Normal	100,000	50,000	400	0.0209	0.0152	No
SF Express	Normal	100,000	50,000	500	0.0167	0.0152	No
SF Express	Normal	100,000	50,000	600	0.0139	0.0152	Yes
SF Express	Normal	100,000	50,000	800	0.0104	0.0152	Yes
SF Express	Hot/peak	200,000	5,000	250	0.0213	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	300	0.0178	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	350	0.0152	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	400	0.0133	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	500	0.0107	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	600	0.0089	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	800	0.0067	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	250	0.0235	0.0214	No
SF Express	Hot/peak	200,000	10,000	300	0.0196	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	350	0.0168	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	400	0.0147	0.0214	Yes

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Brand	Season	Demand/day	Price P (\$)	Throughput q (pph)	Robot UC (\$/parcel)	Human UC seasonal	Unit-cost feasible?
SF Express	Hot/peak	200,000	10,000	500	0.0118	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	600	0.0098	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	800	0.0073	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	250	0.0250	0.0214	No
SF Express	Hot/peak	200,000	13,500	300	0.0209	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	350	0.0179	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	400	0.0156	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	500	0.0125	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	600	0.0104	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	800	0.0078	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	250	0.0279	0.0214	No
SF Express	Hot/peak	200,000	20,000	300	0.0232	0.0214	No
SF Express	Hot/peak	200,000	20,000	350	0.0199	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	400	0.0174	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	500	0.0139	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	600	0.0116	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	800	0.0087	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	250	0.0323	0.0214	No
SF Express	Hot/peak	200,000	30,000	300	0.0269	0.0214	No
SF Express	Hot/peak	200,000	30,000	350	0.0230	0.0214	No
SF Express	Hot/peak	200,000	30,000	400	0.0202	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	500	0.0161	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	600	0.0134	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	800	0.0101	0.0214	Yes
SF Express	Hot/peak	200,000	50,000	250	0.0410	0.0214	No
SF Express	Hot/peak	200,000	50,000	300	0.0342	0.0214	No
SF Express	Hot/peak	200,000	50,000	350	0.0293	0.0214	No
SF Express	Hot/peak	200,000	50,000	400	0.0256	0.0214	No
SF Express	Hot/peak	200,000	50,000	500	0.0205	0.0214	Yes
SF Express	Hot/peak	200,000	50,000	600	0.0171	0.0214	Yes
SF Express	Hot/peak	200,000	50,000	800	0.0128	0.0214	Yes
JD Logistics	Cold/off-peak	50,000	5,000	250	0.0162	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	300	0.0135	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	350	0.0116	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	400	0.0101	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	500	0.0081	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	600	0.0067	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	800	0.0051	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	250	0.0179	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	300	0.0149	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	350	0.0128	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	400	0.0112	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	500	0.0089	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	600	0.0074	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	800	0.0056	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	250	0.0190	0.0182	No
JD Logistics	Cold/off-peak	50,000	13,500	300	0.0159	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	350	0.0136	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	400	0.0119	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	500	0.0095	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	600	0.0079	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	800	0.0059	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	250	0.0212	0.0182	No
JD Logistics	Cold/off-peak	50,000	20,000	300	0.0177	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	350	0.0151	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	400	0.0132	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	500	0.0106	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	600	0.0088	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	800	0.0066	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	250	0.0245	0.0182	No
JD Logistics	Cold/off-peak	50,000	30,000	300	0.0204	0.0182	No
JD Logistics	Cold/off-peak	50,000	30,000	350	0.0175	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	400	0.0153	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	500	0.0123	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	600	0.0102	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	800	0.0077	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	50,000	250	0.0312	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	300	0.0260	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	350	0.0223	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	400	0.0195	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	500	0.0156	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	50,000	600	0.0130	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	50,000	800	0.0097	0.0182	Yes
JD Logistics	Normal	100,000	5,000	250	0.0174	0.0210	Yes
JD Logistics	Normal	100,000	5,000	300	0.0145	0.0210	Yes
JD Logistics	Normal	100,000	5,000	350	0.0124	0.0210	Yes
JD Logistics	Normal	100,000	5,000	400	0.0108	0.0210	Yes
JD Logistics	Normal	100,000	5,000	500	0.0087	0.0210	Yes
JD Logistics	Normal	100,000	5,000	600	0.0072	0.0210	Yes
JD Logistics	Normal	100,000	5,000	800	0.0054	0.0210	Yes
JD Logistics	Normal	100,000	10,000	250	0.0191	0.0210	Yes
JD Logistics	Normal	100,000	10,000	300	0.0159	0.0210	Yes
JD Logistics	Normal	100,000	10,000	350	0.0137	0.0210	Yes
JD Logistics	Normal	100,000	10,000	400	0.0120	0.0210	Yes
JD Logistics	Normal	100,000	10,000	500	0.0096	0.0210	Yes
JD Logistics	Normal	100,000	10,000	600	0.0080	0.0210	Yes
JD Logistics	Normal	100,000	10,000	800	0.0060	0.0210	Yes
JD Logistics	Normal	100,000	13,500	250	0.0204	0.0210	Yes
JD Logistics	Normal	100,000	13,500	300	0.0170	0.0210	Yes
JD Logistics	Normal	100,000	13,500	350	0.0146	0.0210	Yes
JD Logistics	Normal	100,000	13,500	400	0.0127	0.0210	Yes
JD Logistics	Normal	100,000	13,500	500	0.0102	0.0210	Yes
JD Logistics	Normal	100,000	13,500	600	0.0085	0.0210	Yes
JD Logistics	Normal	100,000	13,500	800	0.0064	0.0210	Yes
JD Logistics	Normal	100,000	20,000	250	0.0227	0.0210	No
JD Logistics	Normal	100,000	20,000	300	0.0189	0.0210	Yes
JD Logistics	Normal	100,000	20,000	350	0.0162	0.0210	Yes
JD Logistics	Normal	100,000	20,000	400	0.0142	0.0210	Yes
JD Logistics	Normal	100,000	20,000	500	0.0114	0.0210	Yes
JD Logistics	Normal	100,000	20,000	600	0.0095	0.0210	Yes
JD Logistics	Normal	100,000	20,000	800	0.0071	0.0210	Yes
JD Logistics	Normal	100,000	30,000	250	0.0263	0.0210	No
JD Logistics	Normal	100,000	30,000	300	0.0219	0.0210	No
JD Logistics	Normal	100,000	30,000	350	0.0188	0.0210	Yes
JD Logistics	Normal	100,000	30,000	400	0.0164	0.0210	Yes
JD Logistics	Normal	100,000	30,000	500	0.0131	0.0210	Yes
JD Logistics	Normal	100,000	30,000	600	0.0109	0.0210	Yes
JD Logistics	Normal	100,000	30,000	800	0.0082	0.0210	Yes
JD Logistics	Normal	100,000	50,000	250	0.0334	0.0210	No
JD Logistics	Normal	100,000	50,000	300	0.0278	0.0210	No
JD Logistics	Normal	100,000	50,000	350	0.0239	0.0210	No
JD Logistics	Normal	100,000	50,000	400	0.0209	0.0210	Yes
JD Logistics	Normal	100,000	50,000	500	0.0167	0.0210	Yes
JD Logistics	Normal	100,000	50,000	600	0.0139	0.0210	Yes
JD Logistics	Normal	100,000	50,000	800	0.0104	0.0210	Yes
JD Logistics	Hot/peak	200,000	5,000	250	0.0213	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	300	0.0178	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	350	0.0152	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	400	0.0133	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	500	0.0107	0.0297	Yes

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Brand	Season	Demand/day	Price P (\$)	Throughput q (pph)	Robot UC (\$/parcel)	Human UC seasonal	Unit-cost feasible?
JD Logistics	Hot/peak	200,000	5,000	600	0.0089	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	800	0.0067	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	250	0.0235	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	300	0.0196	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	350	0.0168	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	400	0.0147	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	500	0.0118	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	600	0.0098	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	800	0.0073	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	250	0.0250	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	300	0.0209	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	350	0.0179	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	400	0.0156	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	500	0.0125	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	600	0.0104	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	800	0.0078	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	250	0.0279	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	300	0.0232	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	350	0.0199	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	400	0.0174	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	500	0.0139	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	600	0.0116	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	800	0.0087	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	250	0.0323	0.0297	No
JD Logistics	Hot/peak	200,000	30,000	300	0.0269	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	350	0.0230	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	400	0.0202	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	500	0.0161	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	600	0.0134	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	800	0.0101	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	250	0.0410	0.0297	No
JD Logistics	Hot/peak	200,000	50,000	300	0.0342	0.0297	No
JD Logistics	Hot/peak	200,000	50,000	350	0.0293	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	400	0.0256	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	500	0.0205	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	600	0.0171	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	800	0.0128	0.0297	Yes

Table E9: Simulation_Grid - column block 2

Continuation of Simulation_Grid; first columns are repeated as row identifiers.

Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost (\$)
SF Express	Cold/off-peak	50,000	No	4	10	0.31	641.4
SF Express	Cold/off-peak	50,000	No	7	5	0.66	559.4
SF Express	Cold/off-peak	50,000	Yes	6	5	0.66	511.6
SF Express	Cold/off-peak	50,000	Yes	6	4	0.75	466.6
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	418.8
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	371.1
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	323.4
SF Express	Cold/off-peak	50,000	No	4	10	0.31	661.0
SF Express	Cold/off-peak	50,000	No	7	5	0.66	593.7
SF Express	Cold/off-peak	50,000	No	6	5	0.66	541.0
SF Express	Cold/off-peak	50,000	Yes	6	4	0.75	496.0
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	443.4
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	390.7
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	338.1
SF Express	Cold/off-peak	50,000	No	4	10	0.31	674.7
SF Express	Cold/off-peak	50,000	No	7	5	0.66	617.7
SF Express	Cold/off-peak	50,000	No	6	5	0.66	561.6
SF Express	Cold/off-peak	50,000	No	6	4	0.75	516.6
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	460.5
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	404.5
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	348.4
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	7	5	0.66	662.3
SF Express	Cold/off-peak	50,000	No	6	5	0.66	599.9
SF Express	Cold/off-peak	50,000	No	6	4	0.75	554.8
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	492.4
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	430.0
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	367.5
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	1	13	0.11	657.8
SF Express	Cold/off-peak	50,000	No	6	4	0.75	613.7
SF Express	Cold/off-peak	50,000	No	5	4	0.75	541.4
SF Express	Cold/off-peak	50,000	No	4	4	0.75	469.2
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	396.9
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	2	10	0.31	634.2
SF Express	Cold/off-peak	50,000	No	4	4	0.75	547.7
SF Express	Cold/off-peak	50,000	No	3	4	0.75	455.8
SF Express	Normal	100,000	No	16	12	0.63	1,333
SF Express	Normal	100,000	No	14	11	0.66	1,190
SF Express	Normal	100,000	Yes	12	11	0.66	1,094
SF Express	Normal	100,000	Yes	11	10	0.69	999.2
SF Express	Normal	100,000	Yes	9	10	0.70	903.8
SF Express	Normal	100,000	Yes	7	11	0.66	855.7
SF Express	Normal	100,000	Yes	6	10	0.70	760.6
SF Express	Normal	100,000	No	16	12	0.63	1,411
SF Express	Normal	100,000	No	14	11	0.66	1,258
SF Express	Normal	100,000	No	12	11	0.66	1,153
SF Express	Normal	100,000	Yes	11	10	0.69	1,053
SF Express	Normal	100,000	Yes	9	10	0.70	947.9
SF Express	Normal	100,000	Yes	7	11	0.66	890.0
SF Express	Normal	100,000	Yes	6	10	0.70	790.0
SF Express	Normal	100,000	No	16	12	0.63	1,466
SF Express	Normal	100,000	No	14	11	0.66	1,307
SF Express	Normal	100,000	No	12	11	0.66	1,194
SF Express	Normal	100,000	Yes	11	10	0.69	1,091
SF Express	Normal	100,000	Yes	9	10	0.70	978.8
SF Express	Normal	100,000	Yes	7	11	0.66	914.1
SF Express	Normal	100,000	Yes	6	10	0.70	810.6
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	14	11	0.66	1,396
SF Express	Normal	100,000	No	12	11	0.66	1,271
SF Express	Normal	100,000	No	11	10	0.69	1,161
SF Express	Normal	100,000	Yes	9	10	0.70	1,036
SF Express	Normal	100,000	Yes	7	11	0.66	958.7
SF Express	Normal	100,000	Yes	6	10	0.70	848.8
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	12	11	0.66	1,389
SF Express	Normal	100,000	No	11	10	0.69	1,269

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Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost (\$)
SF Express	Normal	100,000	No	9	10	0.70	1,124
SF Express	Normal	100,000	Yes	7	11	0.66	1,027
SF Express	Normal	100,000	Yes	6	10	0.70	907.7
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	11	10	0.69	1,485
SF Express	Normal	100,000	No	9	10	0.70	1,301
SF Express	Normal	100,000	No	7	11	0.66	1,165
SF Express	Normal	100,000	Yes	6	10	0.70	1,025
SF Express	Hot/peak	200,000	No	32	31	0.60	3,291
SF Express	Hot/peak	200,000	Yes	27	31	0.60	3,053
SF Express	Hot/peak	200,000	Yes	23	31	0.60	2,862
SF Express	Hot/peak	200,000	Yes	20	31	0.60	2,719
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,528
SF Express	Hot/peak	200,000	Yes	14	31	0.60	2,432
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,241
SF Express	Hot/peak	200,000	No	32	31	0.60	3,448
SF Express	Hot/peak	200,000	No	27	31	0.60	3,185
SF Express	Hot/peak	200,000	Yes	23	31	0.60	2,975
SF Express	Hot/peak	200,000	Yes	20	31	0.60	2,817
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,606
SF Express	Hot/peak	200,000	Yes	14	31	0.60	2,501
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,290
SF Express	Hot/peak	200,000	No	32	31	0.60	3,558
SF Express	Hot/peak	200,000	No	27	31	0.60	3,278
SF Express	Hot/peak	200,000	Yes	23	31	0.60	3,053
SF Express	Hot/peak	200,000	Yes	20	31	0.60	2,885
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,661
SF Express	Hot/peak	200,000	Yes	14	31	0.60	2,549
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,325
SF Express	Hot/peak	200,000	No	31	32	0.58	3,757
SF Express	Hot/peak	200,000	No	26	32	0.58	3,444
SF Express	Hot/peak	200,000	No	22	32	0.57	3,195
SF Express	Hot/peak	200,000	Yes	20	31	0.60	3,013
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,763
SF Express	Hot/peak	200,000	Yes	13	32	0.58	2,633
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,388
SF Express	Hot/peak	200,000	No	31	32	0.58	4,061
SF Express	Hot/peak	200,000	No	26	32	0.58	3,699
SF Express	Hot/peak	200,000	No	22	32	0.57	3,410
SF Express	Hot/peak	200,000	No	20	31	0.60	3,209
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,920
SF Express	Hot/peak	200,000	Yes	13	32	0.58	2,760
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,486
SF Express	Hot/peak	200,000	No	1	74	0.0187	4,302
SF Express	Hot/peak	200,000	No	24	35	0.54	4,196
SF Express	Hot/peak	200,000	No	22	32	0.57	3,842
SF Express	Hot/peak	200,000	No	20	31	0.60	3,601
SF Express	Hot/peak	200,000	No	16	31	0.60	3,234
SF Express	Hot/peak	200,000	No	13	32	0.58	3,015
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,683
JD Logistics	Cold/off-peak	50,000	Yes	9	6	0.71	715.1
JD Logistics	Cold/off-peak	50,000	Yes	8	5	0.75	619.8
JD Logistics	Cold/off-peak	50,000	Yes	7	5	0.75	572.1
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	524.3
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	476.6
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	428.9
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	381.1
JD Logistics	Cold/off-peak	50,000	No	9	6	0.71	759.3
JD Logistics	Cold/off-peak	50,000	Yes	8	5	0.75	659.0
JD Logistics	Cold/off-peak	50,000	Yes	7	5	0.75	606.4
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	553.8
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	501.1
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	448.5
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	395.8
JD Logistics	Cold/off-peak	50,000	No	8	6	0.71	790.1
JD Logistics	Cold/off-peak	50,000	No	8	5	0.75	686.5
JD Logistics	Cold/off-peak	50,000	Yes	7	5	0.75	630.4
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	574.4
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	518.3
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	462.2
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	406.1
JD Logistics	Cold/off-peak	50,000	No	9	6	0.71	847.5
JD Logistics	Cold/off-peak	50,000	No	8	5	0.75	737.5
JD Logistics	Cold/off-peak	50,000	No	7	5	0.75	675.0
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	612.6
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	550.2
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	487.7
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	425.3
JD Logistics	Cold/off-peak	50,000	No	1	18	0.0786	928.8
JD Logistics	Cold/off-peak	50,000	No	8	5	0.75	815.9
JD Logistics	Cold/off-peak	50,000	No	7	5	0.75	743.7
JD Logistics	Cold/off-peak	50,000	No	6	5	0.75	671.4
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	599.2
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	526.9
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	454.7
JD Logistics	Cold/off-peak	50,000	No	1	18	0.0786	948.4
JD Logistics	Cold/off-peak	50,000	No	4	12	0.38	938.5
JD Logistics	Cold/off-peak	50,000	No	7	5	0.75	881.0
JD Logistics	Cold/off-peak	50,000	No	6	5	0.75	789.1
JD Logistics	Cold/off-peak	50,000	No	5	5	0.75	697.3
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	605.4
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	513.5
JD Logistics	Normal	100,000	Yes	18	13	0.70	1,510
JD Logistics	Normal	100,000	Yes	15	13	0.70	1,367
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,272
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,176
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,081
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,033
JD Logistics	Normal	100,000	Yes	6	13	0.70	937.6
JD Logistics	Normal	100,000	No	17	14	0.67	1,596
JD Logistics	Normal	100,000	Yes	15	13	0.70	1,441
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,335
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,230
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,125
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,072
JD Logistics	Normal	100,000	Yes	6	13	0.70	967.0
JD Logistics	Normal	100,000	No	17	14	0.67	1,654
JD Logistics	Normal	100,000	Yes	15	13	0.70	1,492
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,380
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,268
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,156
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,100
JD Logistics	Normal	100,000	Yes	6	13	0.70	987.6
JD Logistics	Normal	100,000	No	17	14	0.67	1,763
JD Logistics	Normal	100,000	No	15	13	0.70	1,588
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,463
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,338
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,213

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Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost (\$)
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,151
JD Logistics	Normal	100,000	Yes	6	13	0.70	1,026
JD Logistics	Normal	100,000	No	17	14	0.67	1,930
JD Logistics	Normal	100,000	No	15	13	0.70	1,735
JD Logistics	Normal	100,000	No	13	13	0.70	1,590
JD Logistics	Normal	100,000	No	11	13	0.69	1,446
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,301
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,229
JD Logistics	Normal	100,000	Yes	6	13	0.70	1,085
JD Logistics	Normal	100,000	No	0	42	0	2,104
JD Logistics	Normal	100,000	No	15	13	0.70	2,029
JD Logistics	Normal	100,000	No	13	13	0.70	1,845
JD Logistics	Normal	100,000	No	11	13	0.69	1,662
JD Logistics	Normal	100,000	No	9	13	0.70	1,478
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,386
JD Logistics	Normal	100,000	Yes	6	13	0.70	1,202
JD Logistics	Hot/peak	200,000	Yes	32	40	0.60	3,932
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	3,693
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,502
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,359
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,168
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,073
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	2,882
JD Logistics	Hot/peak	200,000	Yes	32	40	0.60	4,089
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	3,826
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,615
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,457
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,247
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,141
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	2,931
JD Logistics	Hot/peak	200,000	Yes	32	40	0.60	4,199
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	3,918
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,694
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,526
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,302
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,189
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	2,965
JD Logistics	Hot/peak	200,000	No	32	40	0.60	4,403
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	4,090
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,841
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,653
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,404
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,279
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	3,029
JD Logistics	Hot/peak	200,000	No	32	40	0.60	4,716
JD Logistics	Hot/peak	200,000	No	27	40	0.60	4,355
JD Logistics	Hot/peak	200,000	No	23	40	0.60	4,066
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,849
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,560
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,416
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	3,127
JD Logistics	Hot/peak	200,000	No	32	40	0.60	5,344
JD Logistics	Hot/peak	200,000	No	27	40	0.60	4,885
JD Logistics	Hot/peak	200,000	No	23	40	0.60	4,517
JD Logistics	Hot/peak	200,000	No	20	40	0.60	4,242
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,874
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,691
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	3,323

Table E10: Simulation_Grid - column block 3

Continuation of Simulation_Grid; first columns are repeated as row identifiers.

Brand	Season	Demand/day	All-human cost (\$)	Savings %
SF Express	Cold/off-peak	50,000	675.7	0.0508
SF Express	Cold/off-peak	50,000	675.7	0.17
SF Express	Cold/off-peak	50,000	675.7	0.24
SF Express	Cold/off-peak	50,000	675.7	0.31
SF Express	Cold/off-peak	50,000	675.7	0.38
SF Express	Cold/off-peak	50,000	675.7	0.45
SF Express	Cold/off-peak	50,000	675.7	0.52
SF Express	Cold/off-peak	50,000	675.7	0.0217
SF Express	Cold/off-peak	50,000	675.7	0.12
SF Express	Cold/off-peak	50,000	675.7	0.20
SF Express	Cold/off-peak	50,000	675.7	0.27
SF Express	Cold/off-peak	50,000	675.7	0.34
SF Express	Cold/off-peak	50,000	675.7	0.42
SF Express	Cold/off-peak	50,000	675.7	0.50
SF Express	Cold/off-peak	50,000	675.7	0.0014
SF Express	Cold/off-peak	50,000	675.7	0.0858
SF Express	Cold/off-peak	50,000	675.7	0.17
SF Express	Cold/off-peak	50,000	675.7	0.24
SF Express	Cold/off-peak	50,000	675.7	0.32
SF Express	Cold/off-peak	50,000	675.7	0.40
SF Express	Cold/off-peak	50,000	675.7	0.48
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0.0198
SF Express	Cold/off-peak	50,000	675.7	0.11
SF Express	Cold/off-peak	50,000	675.7	0.18
SF Express	Cold/off-peak	50,000	675.7	0.27
SF Express	Cold/off-peak	50,000	675.7	0.36
SF Express	Cold/off-peak	50,000	675.7	0.46
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0.0264
SF Express	Cold/off-peak	50,000	675.7	0.0918
SF Express	Cold/off-peak	50,000	675.7	0.20
SF Express	Cold/off-peak	50,000	675.7	0.31
SF Express	Cold/off-peak	50,000	675.7	0.41
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0.0614

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Brand	Season	Demand/day	All-human cost (\$)	Savings %
SF Express	Cold/off-peak	50,000	675.7	0.19
SF Express	Cold/off-peak	50,000	675.7	0.33
SF Express	Normal	100,000	1,517	0.12
SF Express	Normal	100,000	1,517	0.22
SF Express	Normal	100,000	1,517	0.28
SF Express	Normal	100,000	1,517	0.34
SF Express	Normal	100,000	1,517	0.40
SF Express	Normal	100,000	1,517	0.44
SF Express	Normal	100,000	1,517	0.50
SF Express	Normal	100,000	1,517	0.0700
SF Express	Normal	100,000	1,517	0.17
SF Express	Normal	100,000	1,517	0.24
SF Express	Normal	100,000	1,517	0.31
SF Express	Normal	100,000	1,517	0.38
SF Express	Normal	100,000	1,517	0.41
SF Express	Normal	100,000	1,517	0.48
SF Express	Normal	100,000	1,517	0.0338
SF Express	Normal	100,000	1,517	0.14
SF Express	Normal	100,000	1,517	0.21
SF Express	Normal	100,000	1,517	0.28
SF Express	Normal	100,000	1,517	0.35
SF Express	Normal	100,000	1,517	0.40
SF Express	Normal	100,000	1,517	0.47
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0.0801
SF Express	Normal	100,000	1,517	0.16
SF Express	Normal	100,000	1,517	0.23
SF Express	Normal	100,000	1,517	0.32
SF Express	Normal	100,000	1,517	0.37
SF Express	Normal	100,000	1,517	0.44
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0.0848
SF Express	Normal	100,000	1,517	0.16
SF Express	Normal	100,000	1,517	0.26
SF Express	Normal	100,000	1,517	0.32
SF Express	Normal	100,000	1,517	0.40
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0.0215
SF Express	Normal	100,000	1,517	0.14
SF Express	Normal	100,000	1,517	0.23
SF Express	Normal	100,000	1,517	0.32
SF Express	Hot/peak	200,000	4,324	0.24
SF Express	Hot/peak	200,000	4,324	0.29
SF Express	Hot/peak	200,000	4,324	0.34
SF Express	Hot/peak	200,000	4,324	0.37
SF Express	Hot/peak	200,000	4,324	0.42
SF Express	Hot/peak	200,000	4,324	0.44
SF Express	Hot/peak	200,000	4,324	0.48
SF Express	Hot/peak	200,000	4,324	0.20
SF Express	Hot/peak	200,000	4,324	0.26
SF Express	Hot/peak	200,000	4,324	0.31
SF Express	Hot/peak	200,000	4,324	0.35
SF Express	Hot/peak	200,000	4,324	0.40
SF Express	Hot/peak	200,000	4,324	0.42
SF Express	Hot/peak	200,000	4,324	0.47
SF Express	Hot/peak	200,000	4,324	0.18
SF Express	Hot/peak	200,000	4,324	0.24
SF Express	Hot/peak	200,000	4,324	0.29
SF Express	Hot/peak	200,000	4,324	0.33
SF Express	Hot/peak	200,000	4,324	0.38
SF Express	Hot/peak	200,000	4,324	0.41
SF Express	Hot/peak	200,000	4,324	0.46
SF Express	Hot/peak	200,000	4,324	0.13
SF Express	Hot/peak	200,000	4,324	0.20
SF Express	Hot/peak	200,000	4,324	0.26
SF Express	Hot/peak	200,000	4,324	0.30
SF Express	Hot/peak	200,000	4,324	0.36
SF Express	Hot/peak	200,000	4,324	0.39
SF Express	Hot/peak	200,000	4,324	0.45
SF Express	Hot/peak	200,000	4,324	0.0610
SF Express	Hot/peak	200,000	4,324	0.14
SF Express	Hot/peak	200,000	4,324	0.21
SF Express	Hot/peak	200,000	4,324	0.26
SF Express	Hot/peak	200,000	4,324	0.32
SF Express	Hot/peak	200,000	4,324	0.36
SF Express	Hot/peak	200,000	4,324	0.43
SF Express	Hot/peak	200,000	4,324	0.0051
SF Express	Hot/peak	200,000	4,324	0.0296
SF Express	Hot/peak	200,000	4,324	0.11
SF Express	Hot/peak	200,000	4,324	0.17
SF Express	Hot/peak	200,000	4,324	0.25
SF Express	Hot/peak	200,000	4,324	0.30
SF Express	Hot/peak	200,000	4,324	0.38
JD Logistics	Cold/off-peak	50,000	951.8	0.25
JD Logistics	Cold/off-peak	50,000	951.8	0.35
JD Logistics	Cold/off-peak	50,000	951.8	0.40
JD Logistics	Cold/off-peak	50,000	951.8	0.45

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Brand	Season	Demand/day	All-human cost (\$)	Savings %
JD Logistics	Cold/off-peak	50,000	951.8	0.50
JD Logistics	Cold/off-peak	50,000	951.8	0.55
JD Logistics	Cold/off-peak	50,000	951.8	0.60
JD Logistics	Cold/off-peak	50,000	951.8	0.20
JD Logistics	Cold/off-peak	50,000	951.8	0.31
JD Logistics	Cold/off-peak	50,000	951.8	0.36
JD Logistics	Cold/off-peak	50,000	951.8	0.42
JD Logistics	Cold/off-peak	50,000	951.8	0.47
JD Logistics	Cold/off-peak	50,000	951.8	0.53
JD Logistics	Cold/off-peak	50,000	951.8	0.58
JD Logistics	Cold/off-peak	50,000	951.8	0.17
JD Logistics	Cold/off-peak	50,000	951.8	0.28
JD Logistics	Cold/off-peak	50,000	951.8	0.34
JD Logistics	Cold/off-peak	50,000	951.8	0.40
JD Logistics	Cold/off-peak	50,000	951.8	0.46
JD Logistics	Cold/off-peak	50,000	951.8	0.51
JD Logistics	Cold/off-peak	50,000	951.8	0.57
JD Logistics	Cold/off-peak	50,000	951.8	0.11
JD Logistics	Cold/off-peak	50,000	951.8	0.23
JD Logistics	Cold/off-peak	50,000	951.8	0.29
JD Logistics	Cold/off-peak	50,000	951.8	0.36
JD Logistics	Cold/off-peak	50,000	951.8	0.42
JD Logistics	Cold/off-peak	50,000	951.8	0.49
JD Logistics	Cold/off-peak	50,000	951.8	0.55
JD Logistics	Cold/off-peak	50,000	951.8	0.0241
JD Logistics	Cold/off-peak	50,000	951.8	0.14
JD Logistics	Cold/off-peak	50,000	951.8	0.22
JD Logistics	Cold/off-peak	50,000	951.8	0.29
JD Logistics	Cold/off-peak	50,000	951.8	0.37
JD Logistics	Cold/off-peak	50,000	951.8	0.45
JD Logistics	Cold/off-peak	50,000	951.8	0.52
JD Logistics	Cold/off-peak	50,000	951.8	0.0035
JD Logistics	Cold/off-peak	50,000	951.8	0.0139
JD Logistics	Cold/off-peak	50,000	951.8	0.0743
JD Logistics	Cold/off-peak	50,000	951.8	0.17
JD Logistics	Cold/off-peak	50,000	951.8	0.27
JD Logistics	Cold/off-peak	50,000	951.8	0.36
JD Logistics	Cold/off-peak	50,000	951.8	0.46
JD Logistics	Normal	100,000	2,104	0.28
JD Logistics	Normal	100,000	2,104	0.35
JD Logistics	Normal	100,000	2,104	0.40
JD Logistics	Normal	100,000	2,104	0.44
JD Logistics	Normal	100,000	2,104	0.49
JD Logistics	Normal	100,000	2,104	0.51
JD Logistics	Normal	100,000	2,104	0.55
JD Logistics	Normal	100,000	2,104	0.24
JD Logistics	Normal	100,000	2,104	0.32
JD Logistics	Normal	100,000	2,104	0.37
JD Logistics	Normal	100,000	2,104	0.42
JD Logistics	Normal	100,000	2,104	0.47
JD Logistics	Normal	100,000	2,104	0.49
JD Logistics	Normal	100,000	2,104	0.54
JD Logistics	Normal	100,000	2,104	0.21
JD Logistics	Normal	100,000	2,104	0.29
JD Logistics	Normal	100,000	2,104	0.34
JD Logistics	Normal	100,000	2,104	0.40
JD Logistics	Normal	100,000	2,104	0.45
JD Logistics	Normal	100,000	2,104	0.48
JD Logistics	Normal	100,000	2,104	0.53
JD Logistics	Normal	100,000	2,104	0.16
JD Logistics	Normal	100,000	2,104	0.25
JD Logistics	Normal	100,000	2,104	0.30
JD Logistics	Normal	100,000	2,104	0.36
JD Logistics	Normal	100,000	2,104	0.42
JD Logistics	Normal	100,000	2,104	0.45
JD Logistics	Normal	100,000	2,104	0.51
JD Logistics	Normal	100,000	2,104	0.0829
JD Logistics	Normal	100,000	2,104	0.18
JD Logistics	Normal	100,000	2,104	0.24
JD Logistics	Normal	100,000	2,104	0.31
JD Logistics	Normal	100,000	2,104	0.38
JD Logistics	Normal	100,000	2,104	0.42
JD Logistics	Normal	100,000	2,104	0.48
JD Logistics	Normal	100,000	2,104	0
JD Logistics	Normal	100,000	2,104	0.0355
JD Logistics	Normal	100,000	2,104	0.12
JD Logistics	Normal	100,000	2,104	0.21
JD Logistics	Normal	100,000	2,104	0.30
JD Logistics	Normal	100,000	2,104	0.34
JD Logistics	Normal	100,000	2,104	0.43
JD Logistics	Hot/peak	200,000	5,951	0.34
JD Logistics	Hot/peak	200,000	5,951	0.38
JD Logistics	Hot/peak	200,000	5,951	0.41
JD Logistics	Hot/peak	200,000	5,951	0.44
JD Logistics	Hot/peak	200,000	5,951	0.47
JD Logistics	Hot/peak	200,000	5,951	0.48
JD Logistics	Hot/peak	200,000	5,951	0.52
JD Logistics	Hot/peak	200,000	5,951	0.31
JD Logistics	Hot/peak	200,000	5,951	0.36
JD Logistics	Hot/peak	200,000	5,951	0.39

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Brand	Season	Demand/day	All-human cost (\$)	Savings %
JD Logistics	Hot/peak	200,000	5,951	0.42
JD Logistics	Hot/peak	200,000	5,951	0.45
JD Logistics	Hot/peak	200,000	5,951	0.47
JD Logistics	Hot/peak	200,000	5,951	0.51
JD Logistics	Hot/peak	200,000	5,951	0.29
JD Logistics	Hot/peak	200,000	5,951	0.34
JD Logistics	Hot/peak	200,000	5,951	0.38
JD Logistics	Hot/peak	200,000	5,951	0.41
JD Logistics	Hot/peak	200,000	5,951	0.45
JD Logistics	Hot/peak	200,000	5,951	0.46
JD Logistics	Hot/peak	200,000	5,951	0.50
JD Logistics	Hot/peak	200,000	5,951	0.26
JD Logistics	Hot/peak	200,000	5,951	0.31
JD Logistics	Hot/peak	200,000	5,951	0.35
JD Logistics	Hot/peak	200,000	5,951	0.39
JD Logistics	Hot/peak	200,000	5,951	0.43
JD Logistics	Hot/peak	200,000	5,951	0.45
JD Logistics	Hot/peak	200,000	5,951	0.49
JD Logistics	Hot/peak	200,000	5,951	0.21
JD Logistics	Hot/peak	200,000	5,951	0.27
JD Logistics	Hot/peak	200,000	5,951	0.32
JD Logistics	Hot/peak	200,000	5,951	0.35
JD Logistics	Hot/peak	200,000	5,951	0.40
JD Logistics	Hot/peak	200,000	5,951	0.43
JD Logistics	Hot/peak	200,000	5,951	0.47
JD Logistics	Hot/peak	200,000	5,951	0.10
JD Logistics	Hot/peak	200,000	5,951	0.18
JD Logistics	Hot/peak	200,000	5,951	0.24
JD Logistics	Hot/peak	200,000	5,951	0.29
JD Logistics	Hot/peak	200,000	5,951	0.35
JD Logistics	Hot/peak	200,000	5,951	0.38
JD Logistics	Hot/peak	200,000	5,951	0.44

Optimization G1

Table E11: Optimization_G1 - column block 1

This table reports the optimal robot-human mix at different assumed robot throughputs.

Brand	Season	Demand/day	q (pph)	Robots	Humans	Robot processed	Human processed
SF Express	Cold/off-peak	50,000	300	7	5	33,007	16,993
SF Express	Cold/off-peak	50,000	400	6	4	37,500	12,500
SF Express	Cold/off-peak	50,000	500	5	4	37,500	12,500
SF Express	Normal	100,000	300	14	11	66,014	33,986
SF Express	Normal	100,000	400	11	10	69,158	30,842
SF Express	Normal	100,000	500	9	10	70,000	30,000
SF Express	Hot/peak	200,000	300	27	31	120,000	80,000
SF Express	Hot/peak	200,000	400	20	31	119,454	80,546
SF Express	Hot/peak	200,000	500	16	31	119,454	80,546
JD Logistics	Cold/off-peak	50,000	300	8	5	37,500	12,500
JD Logistics	Cold/off-peak	50,000	400	6	5	37,500	12,500
JD Logistics	Cold/off-peak	50,000	500	5	5	37,500	12,500
JD Logistics	Normal	100,000	300	15	13	70,000	30,000
JD Logistics	Normal	100,000	400	11	13	69,158	30,842
JD Logistics	Normal	100,000	500	9	13	70,000	30,000
JD Logistics	Hot/peak	200,000	300	27	40	120,000	80,000
JD Logistics	Hot/peak	200,000	400	20	40	119,454	80,546
JD Logistics	Hot/peak	200,000	500	16	40	119,454	80,546

Table E12: Optimization_G1 - column block 2

Continuation of Optimization_G1; first columns are repeated as row identifiers.

Brand	Season	Demand/day	Robot share	Daily hybrid cost	All-human workers	All-human cost	Savings %
SF Express	Cold/off-peak	50,000	0.66	617.7	15	675.7	0.0858
SF Express	Cold/off-peak	50,000	0.75	516.6	15	675.7	0.24
SF Express	Cold/off-peak	50,000	0.75	460.5	15	675.7	0.32
SF Express	Normal	100,000	0.66	1,307	32	1,517	0.14
SF Express	Normal	100,000	0.69	1,091	32	1,517	0.28
SF Express	Normal	100,000	0.70	978.8	32	1,517	0.35
SF Express	Hot/peak	200,000	0.60	3,278	76	4,324	0.24
SF Express	Hot/peak	200,000	0.60	2,885	76	4,324	0.33
SF Express	Hot/peak	200,000	0.60	2,661	76	4,324	0.38
JD Logistics	Cold/off-peak	50,000	0.75	686.5	20	951.8	0.28
JD Logistics	Cold/off-peak	50,000	0.75	574.4	20	951.8	0.40
JD Logistics	Cold/off-peak	50,000	0.75	518.3	20	951.8	0.46
JD Logistics	Normal	100,000	0.70	1,492	42	2,104	0.29
JD Logistics	Normal	100,000	0.69	1,268	42	2,104	0.40
JD Logistics	Normal	100,000	0.70	1,156	42	2,104	0.45
JD Logistics	Hot/peak	200,000	0.60	3,918	99	5,951	0.34
JD Logistics	Hot/peak	200,000	0.60	3,526	99	5,951	0.41
JD Logistics	Hot/peak	200,000	0.60	3,302	99	5,951	0.45

Model Formulas

Table E13: Model Formulas*Clean symbolic specification for paper writing.*

Component	Formula	Meaning	Decision use
Robot annual economic cost	$C_R(P) = aP + b$	$a = 0.306$ and $b = 13,362.44$ under the baseline TCO assumptions.	Maps bare robot price to annual economic cost.
Robot annual output	$Q_R(q, s) = q h_R d u_R \theta_R (1 - r_R) \phi_s m_s$	Throughput multiplied by operating hours, workdays, uptime, handling quality, robotable parcel share and seasonal capacity.	Maps robot efficiency to annual good parcels.
Seasonal human unit cost	$UC_{H,j,s} = UC_{H,j} \frac{\lambda_s}{\eta_s}$	Labour cost multiplier divided by the human capacity multiplier.	Makes peak and off-peak labour costs comparable.
Unit-cost break-even	$\frac{C_R(P)}{Q_R(q, s)} \leq UC_{H,j,s}$	Robot unit cost must not exceed human unit cost.	Operational cost parity.
Required robot throughput	$q_{j,s}^{UC}(P) = \frac{C_R(P)}{UC_{H,j,s} h_R d u_R \theta_R (1 - r_R) \phi_s m_s}$	Minimum parcels per hour required for a robot with bare price P .	Answers how efficient the robot must be.
Maximum robot price	$P_{j,s}^{UC}(q) = \frac{UC_{H,j,s} Q_R(q, s) - b}{a}$	Maximum bare robot price allowed at throughput q under unit-cost parity.	Answers how cheap the robot must be.
Three-year payback frontier	$P_{j,s}^{PB}(q) = \frac{T UC_{H,j,s} Q_R(q, s) - T O_F - B}{(1 + \alpha_E + \alpha_S + \alpha_G) + T \alpha_M}$	Bare robot price ceiling under the cash payback target T .	Main investment threshold.
Integer optimisation objective	$\min_{R,H} \left[R \frac{C_R(P)}{d} + H \frac{C_{H,j} \lambda_s}{d} \right]$	Chooses the number of robots R and human workers H that minimise daily operating cost.	Determines the best hybrid robot–human mix.
Capacity constraint	$\min\{RK_R(q, s), \phi_s D_s\} + HK_{H,j,s} \geq D_s$	Robots can only cover the robotable share of demand; humans cover residual demand and exceptions.	Prevents unrealistic full automation.

Sources

Table E14: Sources*All row-wise URLs are retained as plain text to support auditability.*

Input block	Source	How used	Data quality	Limitation
Salary components	Uploaded BOSS/Liepin extraction table	Employer cost, salary components, night shift, formal-worker status, housing/meal/social-insurance indicators	Medium	Recruitment pages are dynamic; figures are public postings rather than internal payroll
Human efficiency	Uploaded efficiency and quality workbook	Human pph assumptions and quality-loss ranges; SF 500 pph benchmark; JD estimates where worker-level data are not public	Medium	Employee-level wrong sorting, missed scanning, damage and rework rates are mostly estimated
Robot prices and TCO	Uploaded humanoid robot 2026 price/BOM/TCO workbook	Robot prices, scanner, software, maintenance, downtime, operator allocation and other TCO items	Medium	Industrial contract prices and maintenance contracts are incomplete
Unitree G1 benchmark	Public robot price in previous robot workbook	Main reference robot with P=\$13,500	High for public price; medium for logistics deployment	Public price does not guarantee industrial parcel-sorting readiness
Seasonal simulation assumptions	Model assumption sheet	Cold/off-peak, normal and hot/peak demand with labour and capacity multipliers	Scenario assumption	Site-level parcel volume must be replaced by real facility data when available